

Hybrid renewable energy systems for seawater-based green hydrogen in Egyptian coastal zones: A case study

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ABSTRACT

As the world accelerates its transition toward net-zero emissions, green hydrogen production via seawater electrolysis presents a promising and innovative pathway for clean energy generation and ensuring long-term energy sustainability. This study investigates the feasibility of green hydrogen production through seawater electrolysis powered by renewable energy sources in four Egyptian coastal cities: New Alamein, El Tor, New Port Said (Salam), and Suez. A hybrid system comprising photovoltaic panels, wind turbines, batteries, a proton exchange membrane electrolyzer, hydrogen storage tanks, and diesel generators is proposed and evaluated under four distinct scenarios. Both stand-alone and grid-connected configurations are evaluated using HOMER Pro software to optimize system design based on techno-economic and environmental criteria. Multiple scenarios, incorporating varying numbers of wind turbines (15, 20, and 50), are assessed for each location. The results are compared depending on energy generation, hydrogen production, storage capacity, excess electricity, and key financial indicators, including net present cost, levelized cost of energy, and levelized cost of hydrogen. The findings indicate that Scenario Four, which achieves the lowest hydrogen production cost of 0.177 \$/kg, demonstrates the most cost-effective performance. This result remains consistent across all studied locations and operating conditions, thereby highlighting the robustness and adaptability of the proposed hybrid system. This research demonstrates the technical viability of integrating renewable energy with seawater electrolysis for sustainable hydrogen production, contributing to Egypt's transition toward a low-carbon energy system and supporting the objectives of its national hydrogen strategy.

ABBREVIATIONS

AEL	Alkaline Water Electrolysis
AES	Annual energy sold to grid
CC	Cycle charging
CI	Clear Index
CO	Carbon monoxide
CO ₂	Carbon dioxide
DG	Diesel Generator
DS	Dispatch strategy
FC	Fuel Cell
H ₂	Hydrogen
HHV _{H₂}	Higher heating value of hydrogen

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IC	Initial cost
LCOE	Levelized Cost of Energy
LCOE _h	Levelized cost of electricity used for hydrogen production
LCOH	Levelized Cost of Hydrogen
LF	Load following
LHV _{diesel}	Higher heating value of diesel fuel
NASA	National aeronautics and space administrative
NCF	Net cash flow
NO	Nitrogen oxides
NPC	Net Present Cost
NPV	Net present value
NREL	National Renewable Energy Laboratory

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OM	Operation and maintenance cost
PEME	Proton Exchange Membrane Electrolysis
PM	Particulate matter
PR	Performance ratio
PV	Photovoltaic
RP	Replacement cost
SMR	Steam Methane Reforming
SO ₂	Sulfur dioxide
SOE	Solid Oxide Electrolysis
STC	Standard Test Condition
STP	Standard Temperature and Pressure
UHC _s	Unburned hydrocarbons
WT	Wind Turbine

Nomenclature

A_{cell}	Electrolyzer cell area
A_{sw}	Swept area by wind turbine blades
A_{US}	Useable area
β_{WT}	Factor of power exponent law
$C_{ann,tot}$	Total project cost per year
C_{des}	Annual cost of water desalination
$C_{des,T}$	Total desalination cost over the project lifetime
$C_{electrolysis}$	Cost of electrolysis process
C_{ez}	Cost of electrolyzer
$C_{ez,ip}$	Cost of electrolyzer input power
C_{Fr}	Cost fraction
CF	electrolyzer capacity factor
C_{H_2O}	Unit cost of desalinated water
$CRF(i, t)$	Capital recovery factor over the project lifetime
C_r	Storage capacity ratio
d	Components degradation factor
$E_{ann,n}$	Electrical energy after n years
$E_{ann,1}$	Electrical energy after one year
$E_{i,activation}$	Electrolyzer activation energy
F	Faraday constant
f	Annual inflation rate
F_{Cr}	Diesel fuel consumption rate
f_{Pr}	Diesel price
g	Gravitational acceleration
h	Proposed height
H_{ez}	Hydrogen production
h_{meter}	Anemometer device height
H_{solar}	Annual solar radiation incident on PV panels
$H_{solar-STC}$	Total solar irradiance at STC
h_{WT}	Wind turbine hub height
i	Real interest rate
$\bar{i}$	Nominal interest rate
I_{max}	Maximum charge current of the battery
J	Surface current density
J_i^{ref}	Current density pre-exponential factor
k	Battery storage rate constant
k_{gen_0}	Fuel curve intercept coefficient
k_{gen_1}	Slope
L	Membrane thickness
m_{diesel}	Diesel mass flow rate
N_{batt}	Batteries number
$P_{batt,ch}$	Maximum charging power stored in the battery
$P_{batt,ch,MCC}$	Maximum charging power stored in the battery at maximum charge current
$P_{batt,ch,MCR}$	Maximum charging power stored in the battery at maximum charge rate
$P_{batt,ch,KIBAM}$	Maximum charging power stored in the battery kinetic battery model
$P_{batt,dch,KIBAM}$	Maximum discharging power of the battery
P_{ez}	Electrolyzer input power
P_{FC}	Fuel cell output power
P_{grid}	The used power from the grid
$P_{gen,output}$	Generator output power
$P_{gen,rated}$	Generator rated power
$P_{inverter}$	Inverter output power
$P_{PV,gen}$	Power generated by photovoltaic panels
$P_{rectifier}$	Rectifier output power
$P_{MPP,STC}$	Photovoltaic panel rated capacity at STC
$P_{WT,gen}$	Wind turbine output power
$P_{WT,gen,STP}$	Power generated by wind turbines at STP

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Q_{H_2}	Annual hydrogen production
Q_{H_2O}	Water consumption for producing 1 kg of hydrogen
R	Universal fluid constant
t	Project lifetime
t_{ez}	Electrolyzer life time
t_{lr}	Temperature lapse rate
T_{ez}	Nominal operating electrolyzer temperature
T_{PVc}	Photovoltaic cell temperature
$T_{PVc,STC}$	photovoltaic cell temperature at STC
T_{STP}	temperature at STP
v_{meter}	Wind speed at anemometer device height
V_{cell}	Electrolyzer cell voltage
$V_{i,activation}$	Electrolyzer activation overpotential
V_{ohm}	Electrolyzer ohmic overpotential
V_{rat}	Rated voltage of the battery
V_{rev}	Electrolyzer reversible overpotential
v_w^3	Wind speed at the propose turbine hub height
W_E	Battery stored energy at the time-step starting
$W_{E,max}$	Total battery stored energy
$W_{E,s}$	Battery available stored energy at the time-step starting
α_c	Maximum charge rate of the battery
γ_p	Temperature coefficient of power
λ_a	Water content at the anode
λ_c	Water content at the cathode
σ_{ez}	Electrolyzer ionic conductivity
ρ	Air real density
ρ_{air}	Air density
η_{batt}	Roundtrip efficiency of the battery
$\eta_{batt,ch}$	Battery charge efficiency
$\eta_{batt,dch}$	Battery discharge efficiency
η_{ez}	Electrolyzer efficiency
η_{gen}	Electrical generator efficiency
$\eta_{inverter}$	Inverter efficiency
η_m	Module efficiency
$\eta_{rectifier}$	Rectifier efficiency
Δt	Time step length

1. Introduction

The accelerating global demand for energy, coupled with growing concerns over climate change and environmental degradation, has intensified the pursuit of sustainable and low-carbon energy solutions. Renewable energy sources (RESs), such as solar, wind, and hydropower, have become central to modern power system development due to their abundance, environmental compatibility, and declining costs. However, to guarantee a sufficient balance between generation and demand, such penetration necessitates significant levels of operational flexibility [1, 2]. These resources are increasingly integrated into standalone and grid-connected systems, enhancing energy security and reducing dependence on fossil fuels. However, the intermittent and location-dependent nature of renewables poses significant challenges to grid stability and energy storage [3,4]. In response, green hydrogen has emerged as a promising energy carrier that can store excess renewable electricity and serve as a versatile fuel for power generation, transportation, and industrial applications [5]. Lotfy et al. [6] present a techno-economic comparison of photovoltaic systems coupled with batteries versus green hydrogen under Cairo's critical-load conditions. Their results reveal that PV/H₂ systems offer substantially lower levelized costs of electricity and shorter payback periods compared to battery configurations—highlighting hydrogen's practical advantages in off-grid or reliability-critical scenarios. Furthermore, Aba et al. [7] demonstrate that hydrogen remains a uniquely viable option for long-duration and hard-to-electrify applications. This paradigm shift underscores the need for integrated renewable-hydrogen systems, particularly in regions with abundant solar and wind potential, such as coastal zones in Egypt similar to the case analysis in Indonesia [8]. The kind of renewable distributed generation, their location, size, and degree of network penetration all affect how severe these side effects are, therefore careful planning and optimization techniques are needed to minimize any potential negative effects while optimizing any positive

ones [9–11].

There has been significant research in recent years on the use of sustainable energy sources for hydrogen production. Several studies have been conducted in Egypt to evaluate the viability and efficiency of different systems. Rami S. El-Emam et al. [12] conducted a comprehensive study on the operation of three scenarios for a grid-connected hydrogen production plant located in El Mostakbal City, Egypt. These scenarios incorporate photovoltaic systems, battery storage, and hydrogen production technologies. The findings indicate that the optimized model successfully meets the load demand, utilizing 64.3 % renewable energy, with a net present cost estimated at \$71.7 million. In a similar vein, Mohamed Nasser et al. [13] proposed a stand-alone hybrid renewable energy system (HRES) for hydrogen production. The study conducted a comprehensive techno-economic analysis of the system, revealing that it achieved competitive production costs: \$3.73 per kilogram of hydrogen and \$0.137 per kilowatt-hour of energy. Additionally, the system demonstrated a significant environmental benefit, with an annual reduction in carbon emissions of approximately 34.47 tons. Furthermore, Hamdy Hassan et al. [14] utilized MATLAB Simulink to conduct a comparative analysis between a stand-alone hybrid photovoltaic/hydrogen storage system and a conventional photovoltaic/battery system. The findings demonstrate that the electric system exhibits high efficiency. Furthermore, although the hydrogen-based system proves to be more economical, it also contributes to a substantial reduction in carbon dioxide emissions, amounting to 25.6 tons annually.

On the other hand, Hammou Tebibel [15] conducted a study on a wind-based hydrogen production system in Algeria. The study demonstrated that the integration of batteries significantly enhanced the system's performance under low wind speed conditions. This approach resulted in a levelized cost of hydrogen of \$33.70 per kilogram and an annual reduction of 87.75 tons of CO₂ emissions. Similarly, Manaf Zaghaibeh et al. [16] investigated the feasibility of utilizing a photovoltaic hydrogen station for the production of green hydrogen in Oman. The system comprises 5 MW of photovoltaic panels, an electrolyzer, a converter, and hydrogen storage tanks. The findings indicated that the system is capable of producing 90,910 kg of hydrogen annually, with a levelized hydrogen energy cost of 7.23 \$/kg. A study conducted in Chile by Matias Garcia G. et al. [17] examined the performance of a

stand-alone PV/WT system for the production of green hydrogen. The analysis revealed that hybrid electrolysis exhibits greater efficiency compared to traditional methods. The findings indicated that the system achieved a levelized cost of hydrogen of \$2.09 per kilogram, with a carbon dioxide emissions rate of 1.06 kg per kilogram of hydrogen. Finally, Ilham Ourya et al. [18] conducted a comprehensive study in Morocco to develop a large-scale green hydrogen production system. This system is primarily powered by RESs, incorporating storage tanks for hydrogen storage. The findings indicate that the hybrid system, which utilizes storage tanks in Dakhla, achieves the lowest production cost, amounting to \$2.54 per kilogram. Based on the prior literature review, Table 1 presents a comparative analysis between the proposed study and existing research, illustrating the gradient of the hydrogen production system. This study aims to explore the optimal design of a grid-connected hybrid renewable energy system for efficient hydrogen production. The hydrogen generated can serve multiple purposes, including generating electricity, powering hydrogen-based loads such as electric vehicles (EVs), or being stored in tanks for later use. This stored hydrogen has the potential to be transported to regions with limited access to water and renewable energy resources, addressing critical energy needs. Furthermore, hydrogen can be exported through maritime infrastructure in coastal cities, maximizing commercial profitability and driving the development of a sustainable hydrogen economy. In recognition of these considerations, Egypt's National Strategy for Low-Carbon Hydrogen—officially approved by the Supreme Council of Energy on 27 February 2024—established a comprehensive framework for scaling hydrogen production for both domestic applications (industry and transport) and export, under the oversight of the National Council for Green Hydrogen [19]. Moreover, a landmark offtake agreement concluded under Germany's H2Global pilot auction commits the Egypt Green Hydrogen consortium—comprising Fertigllobe, Scatec, Orascom, the Sovereign Fund of Egypt, and the Egyptian Electricity Transmission Company—to export up to 397,000 tons of green ammonia, produced from Egyptian green hydrogen, between 2027 and 2033 at a fixed price of 1165.38 \$/ton [20]. The proposed system not only enhances national energy security and reduces dependence on fossil fuels but also aligns with global efforts to mitigate greenhouse gas emissions and promote environmental sustainability.

Egypt hosts COP 27 to combat climate change and promote energy

Table 1
Summary of literature review.

Ref No, Year	Technology	System Type	Objective of research	Methodology	Key finding	LCOH (\$/kg)	CO ₂ reduction (Tons/year)
[12] 2022	GC	PV/Battery	Evaluates three scenarios of green electrical energy demand	Hybrid optimization using HOMER	Encourage the role of renewable energy for power generation	–	Produce: 975.64
[13] 2022	SA	PV/WT	Assess of a hydrogen production system from HRES	Energy and Exergy Analysis using MATLAB	Performs exergetically assessment of a green H ₂ production	3.73	34.47
[14] 2023	SA	PV	Assess the performance of a streetlighting PV/H ₂ system	Comparison among storage systems using MATLAB	The PV/H ₂ system is more economical than PV/Battery system	–	25.6
[15] 2021	SA	WT/Battery	Performs multi-objective optimization of WT/H ₂ system	Multi-objective optimization for green H ₂ production	Enhances power and hydrogen management strategy	33.70	87.75
[16] 2022	SA	PV	Discusses the feasibility of the conversion of solar energy into hydrogen	Analytical model for a techno-economic H ₂ production	Improved performance of system components cost	7.26	–
[17] 2023	SA	PV/WT	Evaluates H ₂ production and CO ₂ emissions from PV/WT system	Linear optimization model to design the hybrid green H ₂ plant	Indicates that the costs were competitive in the market	2.09	Produce: 1.57 kg CO ₂ e/kg H ₂
[18] 2023	SA	PV/WT/Battery	Performs a technical-economic analysis of H ₂ production from WT/PV system	Optimization of autonomous hybrid systems using HOMER	Choosing the best site for hydrogen in Morocco	2.54	–
Proposed	GC	PV/WT/DG/Battery	Minimization of net present cost and gas emissions for H ₂ production	Multi-objective optimization Using HOMER PRO	Achieves net cost and emissions minimizations	0.177:2.95	100 % RF

GC: Grid connected, SA: Stand-alone, RF: Renewable fraction.

sustainability through green economy projects, renewable energy diversification, and implementing the Paris Agreement. The Sustainable Energy Strategy for 2035 aims to boost renewable energy in Egypt to 42 % of electricity production by 2035, with 7000 MW installed. New projects include green hydrogen production in Ain Sokhna and East Port Said. The national green hydrogen strategy aims to decrease the cost of green hydrogen production from 2.7 \$/kg in 2025 to 1.7 \$/kg in 2050 [21]. The paper conducts a technical-environmental-economic study on the optimal sizing of microgrid components for green hydrogen production in Egypt, using the Hybrid Optimization Model for Electric Renewables (Homer Pro) software, considering factors like work environment, economic conditions, load characteristics, and available energy sources.

The load window outlines various loads, including electric, thermal, and hydrogen, and profiles like residential, commercial, industrial, and community. It includes components like fuel cells (FCs), storage batteries, converters, and reformers, analyzing economic variables to determine the optimal design. The proposed application is implemented via four stages including simulation, optimization, sensitivity analysis, and optimal selection. It analyzes the project's techno-environmental-economic aspects, determining the system with the lowest cost and highest revenue, considering factors like net present cost (NPC), levelized cost of energy (LCOE), levelized cost of hydrogen (LCOH), energy, hydrogen, renewable penetration, and CO₂ emissions [22]. The primary contributions of this manuscript are summarized as follows:

- Proposing cost-effective microgrid designs tailored to hydrogen production in newly developed urban areas, thereby promoting sustainable infrastructure development.
- Enhancing system cost-efficiency by allowing the sale of surplus electricity through grid connectivity, thereby generating additional revenue and improving economic sustainability.
- Integrating techno-economic, environmental, and sensitivity analyses to evaluate financial performance and decarbonization potential, considering twelve scenarios and balancing short-term and long-term storage options.
- Quantifying the impacts of turbine selection and local wind regimes on hydrogen yield, LCOE, and system economics across four coastal

cities, providing insights into location-dependent techno-economic and environmental outcomes.

2. Overview of the system

2.1. Location information

The study investigates the production of green hydrogen for supplying electrical energy to the residential compound “Dar Misr” in four Egyptian cities. The proposed green hydrogen plants are expected to be installed in New Alamein (on the Mediterranean Sea) at an elevation of 7 m (30° 51.3' N, 28° 56.7' E), El Tor (on the Red Sea) at an elevation of 14 m (28° 15.0' N, 33° 36.2' E), New Port Said “Salam” (on the Mediterranean Sea) at an elevation of 6 m (31° 14.6' N, 32° 18.0' E), and Suez (on the Red Sea) at an elevation of 11 m (29° 57.3' N, 32° 31.1' E) [23]. Fig. 1 illustrates the proposed location for each city from Google Maps.

2.2. Load description

The study sites, New Alamein, El Tor, New Port Said ‘Salam’, and Suez, are determined by traditional domestic load patterns for 10 middle-income housing buildings. Each building has 6 floors, and each floor contains 4 residential units. There are 3 models for the areas of residential units in each building: 10 units with an area of 110 m², 10 units with an area of 130 m², and 4 units with an area of 150 m² [23].

The system load description is provided in Appendix A as follows. The annual electricity demand for the study sites in 2024 was calculated based on the loads summarized in Table A1. The estimated daily load for each apartment model during different seasons (spring/summer and fall/winter) is presented in Table A2. The total daily energy consumption for each building across seasons is reported in Table A3. The load characteristics, including annual usage, average load, peak load, and load factor, are summarized in Table A4. Finally, the total estimated daily hydrogen and electricity demands for the proposed site under different seasonal conditions are provided in Table A5.

During Summer, the load consumes a low energy rate from 7 a.m. to 3 p.m. while the load is maximum from 6 p.m. to 8 p.m. daily. On the other hand, the load consumes a low energy rate from 11 p.m. to 3 p.m.



Fig. 1. Geographic locations of the proposed case studies.

Table 2

Technical characteristics and specifications of the system components [29,31, 35].

Unit	Description	Unit	Value
Generic flat plate PV	Rated capacity for each PV module	kW	4000
	Temperature coefficient	%/°C	−0.5
	Nominal operating cell temperature	°C	47
	Efficiency at STC	%	13
	Ground reflection	%	20
Windflow 45 WT	Derating factor	%	80
	Rated capacity for turbine	kW	500
	Rotor diameter	M	45
	Specific area	m ² /kW	3.19
	Voltage	V	415–480
	Number of blades	–	2
	Minimum/Actual/Maximum Hub height	M	31,5/38/47
	Cut-in/Rated/Cut-off wind speed	m/s	4,5/14/25
	Rated voltage	V	600
Generic 1 MWh Li-ion	Nominal capacity	kWh	1E+03
	Maximum capacity	Ah	1.67E+03
	Roundtrip efficiency	%	90
	Maximum charge current	A	1.67E+03
	Maximum discharge current	A	5E+03
	Throughput	kWh	3E+06
	Minimum state of charge	%	20
	Lifespan	Years	10
	Minimum load ratio	%	25
	Efficiency	%	66
	Operating temperature	°C	50–80
	Lifespan	Years	10
	Rated capacity	kW	10
Generic PEM Electrolyzer	Fuel curve intercept coefficient	L/hr/kW	0.0480
	Fuel curve slope	L/hr/kW	0.2860
	Minimum load ratio	%	25
Generic electric converter	Relative capacity	%	100
	Conversion efficiency	%	95

Table 3

Operating and technical parameters.

Parameter	Unit	Value
Real discount rate	%	21.25
Inflation rate	%	33.70
Project lifetime	Years	25
Grid power price	\$/kWh	0.1
Grid sell back price	\$/kWh	0.05

during winter. Fig. 2 shows the daily load profile for the proposed sites. The load characteristics are described by the following: annual usage of 1.257 GWh/yr, average load of 143.49 kW, peak load of 926.56 kW, and a load factor of 0.15.

2.3. Meteorological data

The National Aeronautics and Space Administration (NASA) collects various climate data required to determine the suitable RESs for four study sites (New Alamein, El Tor, New Port Said “Salam”, and Suez) [24]. The data includes solar radiation, clear index (CI), wind speed, and temperature measurements [25]. It assesses the feasibility of WTs and calculates the impact on solar panels, meeting 0.4 % of global energy demand [26].

The meteorological inputs (solar radiation, wind speed, and temperature) were obtained as monthly averages covering the time span 1984–2024 at a 0.5° × 0.5° spatial resolution, consistent with the data availability in HOMER Pro. These data for the various case studies are

Table 4

Financial details of the proposed system components [29,38].

Parameter	Capital cost (\$)	Replacement cost (\$)	Operating and maintenance cost (\$)	Lifetime
Solar system	2500/1 kW	2500/1 kW	10/yr	25 yr
Wind system	500,000/turbine	500,000/turbine	100/yr	20 yr
Battery storage system	150,000/battery	150,000/battery	10,000/yr	10 yr
PEM electrolyzer	2500/unit	2500/unit	1/yr	10 yr
Backup generator	5000/10 kW	5000/10 kW	0.300/operating hour	15,000 h
Hydrogen tank	900/tank	900/tank	50/yr	25 yr
Hydrogen FC	1280/kW	1280/kW	0.05/kW/yr	90,000 h
Electric converter	300/kW	300/kW	20/yr	15 yr

presented in Fig. 3. Between April and September, New Alamein City exhibits high solar irradiance, with an average CI of 0.68. Wind speeds range from 5.14 m/s in November to 7.23 m/s in January, with an annual average of 6.02 m/s, as illustrated in Fig. 3a. Additionally, the city experiences elevated temperatures from June to October, with an average of 20.56 °C. El Tor City (Fig. 3b) experiences high solar irradiance, wind speeds, and temperatures from March to September, with an average CI of 0.70, and wind speeds varying from 4.25 to 6.00 m/s with higher temperatures during the period from June to September, averaging 20.4 °C.

New Port Said “Salam” City (Fig. 3c) experiences high solar irradiance from April to September, with an average CI of 0.62. Wind speeds range from 4.48 to 6.70 m/s, with an annual average of 5.38 m/s while elevated temperatures are also observed between July and September, with an average of 21.48 °C. Suez City experiences substantial solar irradiance between March and September, with an average CI of 0.70. Wind speeds range from 4.73 m/s in November to 6.15 m/s in March, with an annual average of 5.35 m/s, as depicted in Fig. 3d. High temperatures are recorded from June to September, with an average of 21.46 °C.

3. Methodology

The proposed case studies utilize Homer Pro software for simulated hydrogen and electrical loads, with Fig. 4 illustrating the microgrid design models’ optimization framework. The system’s objective functions, constraints, and technical characteristics are discussed.

3.1. Objective function formulation

The study examines stand-alone and grid-connected hybrid systems (PV/WT/Battery/PEME and PV/WT/DG/Battery/PEME) for electricity and hydrogen production, presenting mathematical formulations for NPC in different configurations as presented through equations (1)–(4). In addition, equation (5) represents the objective of minimization of gas emissions [27].

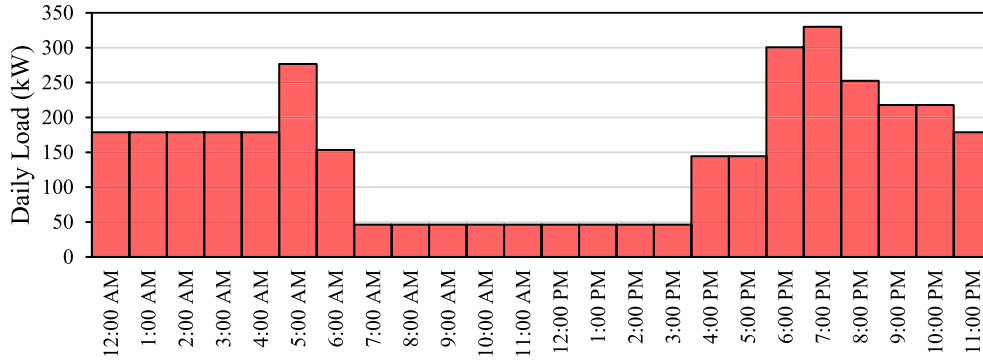
$$F_{NPC_1} = \sum_{i=1}^{N_i} NPC_i^{PV} + \sum_{j=1}^{N_j} NPC_j^{WT} + \sum_{k=1}^{N_k} NPC_k^{batt} + \sum_{l=1}^{N_l} NPC_l^{H_2} + pf_e \quad (1)$$

$$F_{NPC_2} = \sum_{i=1}^{N_i} NPC_i^{PV} + \sum_{j=1}^{N_j} NPC_j^{WT} + \sum_{k=1}^{N_k} NPC_k^{batt} + \sum_{l=1}^{N_l} NPC_l^{H_2} + \sum_{m=1}^{N_m} AES_m^{grid} + pf_e \quad (2)$$

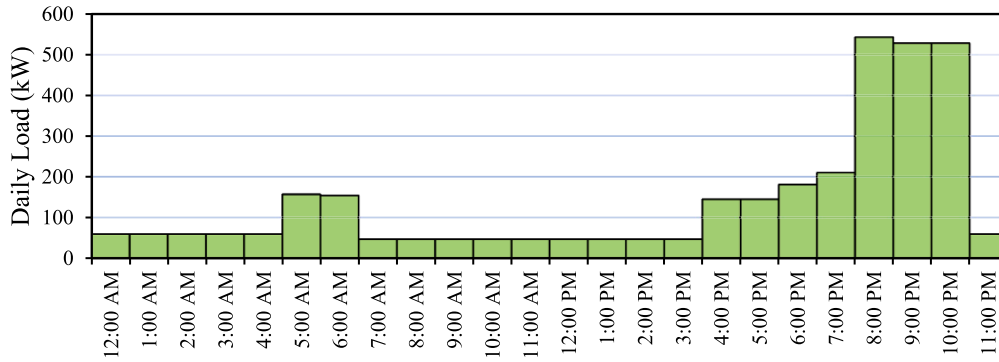
Table 5Energy analysis of a stand-alone PV/WT/Battery/PEME/H₂ tank system for each site studied.

WT		PV/WT/Battery/PEME/H ₂ tank								
		Energy (kWh/yr)					Hydrogen (kg/yr)		Autonomy (hr)	
		PV	WT	Battery	AC load	Shortage	PEME	H ₂ load	H ₂ tank	Battery
New Alamein	15	13,031,123	21,883,028	340,261	1,256,452	892	44,524	44,286	92.9	22.3
	20	5,114,000	29,177,371	334,741	1,256,398	1071	44,534	44,285	92.9	22.3
	50	1,877,595	72,943,426	317,978	1,256,250	1253	44,583	44,317	92.9	22.3
El Tor	15	–	–	–	–	–	–	–	–	–
	20	16,390,562	25,272,174	346,672	1,256,887	193	44,593	44,285	92.9	22.3
	50	5,044,255	63,180,435	325,241	1,256,806	646	44,598	44,285	92.9	22.3
Salam	15	16,825,337	18,185,530	387,291	1,256,286	1188	44,542	44,285	92.9	22.3
	20	8,877,310	24,247,373	380,937	1,256,950	0	44,550	44,285	92.9	27.9
	50	2,830,222	60,618,433	360,623	1,256,950	0	44,556	44,285	92.9	27.9
Suez	15	13,830,210	19,737,120	365,678	1,256,297	1029	44,507	44,285	92.9	22.3
	20	7,035,474	26,316,159	358,189	1,256,299	1179	44,524	44,285	92.9	22.3
	50	3,003,696	65,790,398	335,291	1,256,230	1252	44,551	44,285	92.9	22.3

(a) Spring - Summer seasons



(b) Fall - Winter seasons

**Fig. 2.** A daily electric load profile sampled across various seasons.

$$F_{NPC_3} = \sum_{i=1}^{N_i} NPC_i^{PV} + \sum_{j=1}^{N_j} NPC_j^{WT} + \sum_{k=1}^{N_k} NPC_k^{batt} + \sum_{l=1}^{N_l} NPC_l^{H_2} + \sum_{R=1}^{N_R} NPC_R^{gen} + pf_e \quad (3)$$

$$F_{NPC_3} = \sum_{i=1}^{N_i} NPC_i^{PV} + \sum_{j=1}^{N_j} NPC_j^{WT} + \sum_{k=1}^{N_k} NPC_k^{batt} + \sum_{l=1}^{N_l} NPC_l^{H_2} + \sum_{R=1}^{N_R} NPC_R^{gen} + \sum_{m=1}^{N_m} AES_m^{grid} + pf_e \quad (4)$$

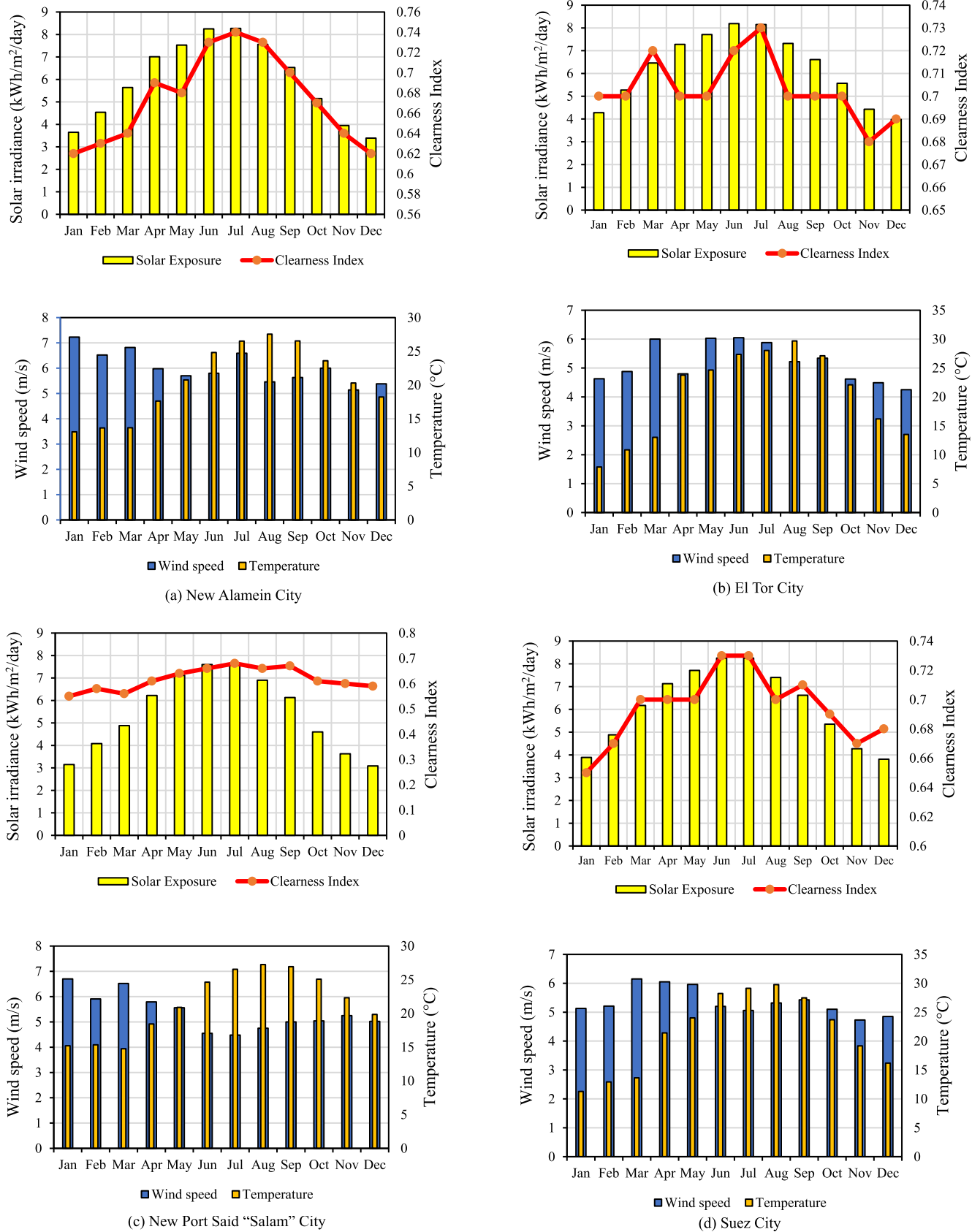


Fig. 3. Meteorological data for different case studies.

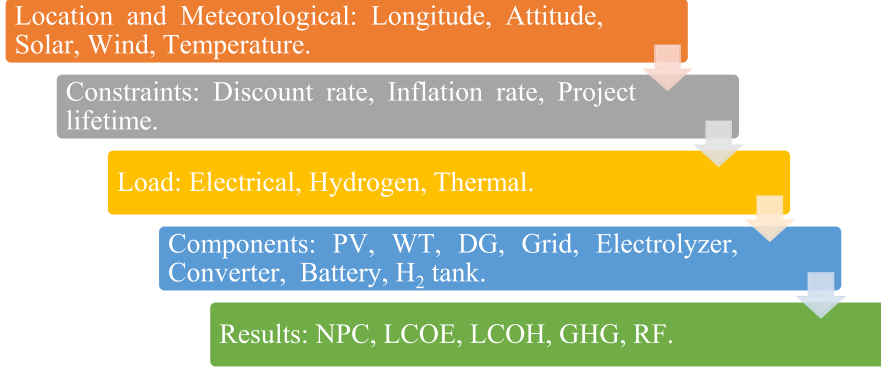


Fig. 4. Implemented optimization framework.

$$F_{emiss}^{gen} = \sum_{R=1}^{N_R} E_R(P_{gen_R}) = \sum_{R=1}^{N_R} (a_R P_{gen}^2 + b_R P_{gen} + c_R) + x_R e^{y_R P_{gen_R}} \quad (5)$$

where, pf_e is the penalty function ensuring operating variables remain within specified ranges during optimization. Also, a_r , b_r , c_r , x_r , and y_r represent generator emission coefficients.

3.2. Operational constraints

The objective functions are influenced by equality and inequality conditions, based on the specific configurations of the hybrid system [28], as illustrated in Equations (6) and (7) considering both the real and reactive power loads.

$$\sum_i^{N_i} P_{PV,gen_i} + \sum_j^{N_j} P_{WT,gen_j} = P_{load} + P_{loss} \quad (6)$$

$$\sum_i^{N_i} Q_{PV,gen_i} + \sum_j^{N_j} Q_{WT,gen_j} = Q_{load} + Q_{loss} \quad (7)$$

Equation (8) through (11) outline inequality constraints, defining maximum output power limits for solar panels, limitations on WT's active and reactive output power, and permissible variation for battery state of charge, based on irradiance and temperature.

$$P_{PV,gen_i} \leq P_{PV,gen_i}^{max} \quad (8)$$

$$P_{WT,gen_j} \leq P_{WT,gen_j}^{max} \quad (9)$$

$$Q_{WT,gen_j}^{min} \leq Q_{WT,gen_j} \leq Q_{WT,gen_j}^{max} \quad (10)$$

$$SOC_k^{batt,min} \leq \left[(1 - dis^{batt}) SOC_k^{batt,0} - \Delta t \frac{\eta_{batt}}{W_E} \sum_{k=1}^k P_{batt, ch}^k \right] \leq SOC_k^{batt,max} \quad (11)$$

3.3. Components of the proposed system

The proposed system may be a combination of PV, WT, battery, electrolyzer, diesel generator (DG), hydrogen tank, FC, and electric converter. This subsection discusses the system components, description, and specifications, with Fig. 5 showcasing the proposed system components.

3.3.1. Solar system

Photovoltaic Solar modules convert incident global radiation into electric energy to provide the energy needed by water electrolyzer for hydrogen production. The power generated by solar panels depends on the direct normal and horizontal incident solar radiation, the cell

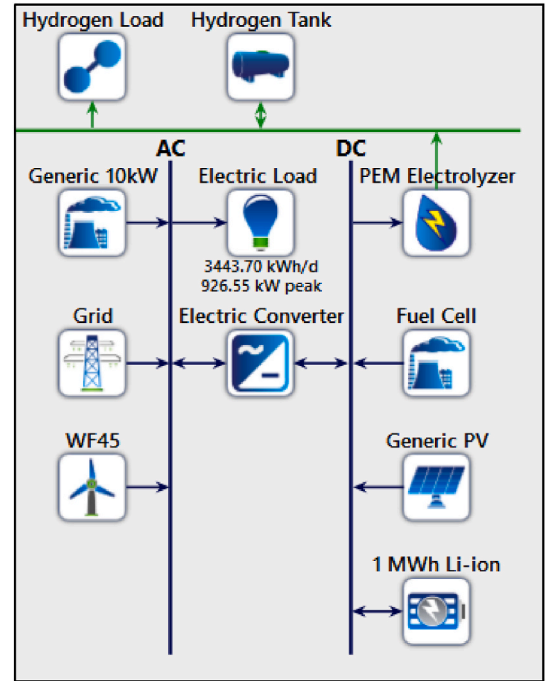


Fig. 5. Proposed system components.

temperature, and the performance ratio (derating factor) which refers to the power loss due to dust, reflection, bird droppings, line losses, and other environmental conditions. The power generated by photovoltaic panels can be calculated using the following equations [29]:

PV solar modules convert global radiation into electric energy to provide the energy needed by water electrolyzer for hydrogen production. The power generated depends on solar radiation, cell temperature, and performance ratio, which includes power loss due to dust, reflection, bird droppings, line losses, and other environmental conditions. The power generated by solar panels are calculated as follows [29]:

$$P_{PV,gen} = P_{MPP,STC} PR \left(\frac{H_{solar}}{H_{solar-STC}} \right) \left[1 + \gamma_p (T_{PVC} - T_{PVC,STC}) \right] \quad (12)$$

$$P_{MPP} = A_{US} \eta_m H_{solar} \quad (13)$$

where $P_{MPP,STC}$ is the photovoltaic panel rated capacity (kW) at standard test condition (STC), A_{US} is the useable area m^2 , η_m is the module efficiency, PR (%) is the PV performance ratio that ranges from 0.5 (very poor system) to 0.85 (top-quality system), H_{solar} is the annual solar radiation incident on PV panels (kW/m^2), $H_{solar-STC}$ is the total irradiance at standard conditions, γ_p is the temperature coefficient of power which

represents the adverse effect of increasing PV temperature on power ($\%/^{\circ}\text{C}$), T_{PVc} is the cell temperature ($^{\circ}\text{C}$), and $T_{PVc,STC}$ is the cell temperature under reference conditions. Generic flat-plate PV panels with different panel slopes according to different sites without a tracking system are used. Table 2 presents the technical details of the proposed solar system at standard test conditions (Cell temperature $T_{PVc,STC} = 25^{\circ}\text{C}$, Solar irradiance $H_{solar-STC} = 1 \text{ kW/m}^2$, Air mass = 1.5).

3.3.2. Wind system

WTs convert wind energy into electric energy, sharing photovoltaic energy for hydrogen production. Their output power depends on factors like air density, swept area, wind speed, rotor altitude, hub height, and ambient temperature. [30]. WTs operate between cut-in and cut-out speeds, reducing rotation speed with low or high wind speed to optimize wind use without damaging the turbine. Power generation and actual power can be calculated using specific equations [29]:

$$P_{WT,gen,STP} = \frac{1}{2} \rho_{air} A_{sw} v_w^3 \quad (14)$$

$$v_w = v_{meter} \left(\frac{h_{WT}}{h_{meter}} \right)^{\beta_{WT}} \quad (15)$$

$$P_{WT,gen} = P_{WT,gen,STP} \frac{\rho}{\rho_{air}} \quad (16)$$

$$\frac{\rho}{\rho_{air}} = \left(\frac{T_{STP}}{T_{STP} - t_{lr} \cdot h} \right) \left(\frac{1 - t_{lr} \cdot h}{T_{STP}} \right)^{\frac{g}{R \cdot t_{lr}}} \quad (17)$$

At STP, $P_{WT,gen,STP}$ is the WT-generated power (kW), ρ_{air} is air density (1.225 kg/m^3). Also, A_{sw} is the swept area by WT-blades (m^2), v_w is the wind speed at the proposed turbine hub height (m/s), v_{meter} is the wind speed at anemometer device height (m/s), h_{WT} is the hub height (m), h_{meter} is the anemometer device height (m/s), and β_{WT} is the factor of power exponent law. At actual conditions, $P_{WT,gen}$ is the WT-output power (kW), and ρ is the air real density (kg/m^3). Also, $\frac{\rho}{\rho_{air}}$ is the air density ratio, T_{STP} is the temperature at standard temperature and pressure (15.01°C), t_{lr} is the temperature lapse rate ($^{\circ}\text{C/m}$), h is the proposed height (m), g is the gravitational acceleration (m/s^2) and R is the universal fluid constant ($1.051 \text{ J/kg}^{\circ}\text{C}$).

With respect to wind resource assessment, HOMER Pro applies the logarithmic wind profile law to adjust the NASA POWER wind speeds (available at 50 m) to the hub heights of the selected turbines. The corresponding height adjustment curves have now been included in Appendix B.

The Windflow 45 WT model is used with specifications given in Table 2. It should be clarified that the reported ratings (4000 kW PV and 500 kW WT) do not represent the initial installed size of a single residential installation, but rather the upper capacity limits defined in HOMER Pro for system optimization. The software subsequently selects the most techno-economically optimal subset of these resources during

the simulation. The relation between wind speeds and turbine output power at the proposed hub height is illustrated through the WT power-speed curve shown in Fig. 6 [31].

3.3.3. Battery storage system

Storage devices, such as batteries, provide energy for load operation during disturbances in primary source. However, batteries represent a financial burden on energy production projects as they require a large economic cost [32]. The battery charging and discharging capacity can be calculated as follows [29,33]:

$$P_{batt,ch} = \frac{\text{Min}(P_{batt,ch,MCC}, P_{batt,ch,MCR}, P_{batt,ch,KiBAM})}{\eta_{batt,ch}} \quad (18)$$

$$P_{batt,ch,MCC} = \frac{N_{batt} \times V_{rat} \times I_{max}}{1000} \quad (19)$$

$$P_{batt,ch,MCR} = \frac{(W_{E,max} - W_E)(1 - e^{-\alpha_c \Delta t})}{\Delta t} \quad (20)$$

$$P_{batt,ch,KiBAM} = \frac{kW_{E,s}e^{-k\Delta t} + W_E k C_r (1 - e^{-k\Delta t})}{1 - e^{-k\Delta t} + C_r(k\Delta t - 1 + e^{-k\Delta t})} \quad (21)$$

$$P_{batt,dch,KiBAM} = \frac{-kC_r W_{E,max} + kW_{E,s}e^{-k\Delta t} + kC_r W_E (1 - e^{-k\Delta t})}{1 - e^{-k\Delta t} + C_r(k\Delta t - 1 + e^{-k\Delta t})} \quad (22)$$

where $P_{batt,ch}$ is the maximum charging power stored in the battery (kW), and $\eta_{batt,ch}$ is the charging efficiency of the battery (%). $P_{batt,ch,MCC}$, $P_{batt,ch,MCR}$, and $P_{batt,ch,KiBAM}$ are the maximum charging power stored in the battery (kW) at maximum charge current, maximum charge rate, and kinetic battery model, respectively. Also, N_{batt} is the battery number, V_{rat} is the rated voltage of the battery (V), I_{max} is the maximum charge current of the battery (A), W_E is the battery stored energy at the time-step starting (kWh), $W_{E,max}$ is the total battery stored energy (kWh), α_c is the maximum charge rate of the battery (A/Ah), Δt is the time step length (h), k is the battery storage rate constant ($1/\text{h}$), $W_{E,s}$ is the battery available stored energy at the time-step starting (kWh), C_r is the storage capacity ratio, and $P_{batt,dch,KiBAM}$ is the maximum discharging power of the battery (kW). The battery lifespan ranges from 5 to 10 years. The Generic 1 MWh lithium-ion battery model is used. The proposed battery specifications are given in Table 2.

3.3.4. PEM electrolyzer

An electrolyzer is a device that uses electricity to split water into its elements of hydrogen and oxygen molecules. It consists of three parts: anode, cathode, and diaphragm. The anode is the positive electrode that attracts oxygen molecules. The cathode is the negative electrode that attracts hydrogen molecules. The diaphragm separates the anode from the cathode to prevent the water from reforming. The electrolyzer input power can be calculated according to the following equations [29,30]:

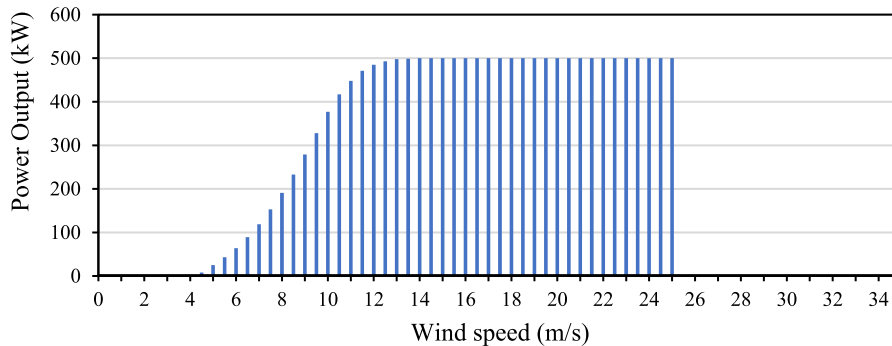


Fig. 6. Windflow 45 WT power-speed curve.

$$P_{ez} = J \times A_{cell} \times V_{cell} \quad (23)$$

$$V_{cell} = V_{rev} \times V_{i,activation} \times V_{ohm}, i : anode \text{ and } cathode \quad (24)$$

$$V_{rev} = 1.229 - [8.5 \times 10^{-4} \times (T_{ez} - 298)] \quad (25)$$

$$V_{i,activation} = \frac{RT_{ez}}{F} \sinh^{-1} \left[\frac{J}{2J_i} \right], J_i = J_i^{ref} e^{-\frac{E_{i,activation}}{RT_{ez}}} \quad (26)$$

$$V_{ohm} = J \int_0^L \frac{dx}{\sigma_{ez}[\lambda(x)]}, \lambda(x) = \frac{\lambda_a - \lambda_c}{L} x + \lambda_c \quad (27)$$

$$\sigma_{ez}[\lambda(x)] = [0.5139\lambda(x) - 0.326] e^{1268 \left(\frac{1}{303} - \frac{1}{T_{ez}} \right)} \quad (28)$$

$$\eta_{ez} \% = \frac{HHV_{H_2}}{H_2 \text{ energy production}} \quad (29)$$

where J is the surface current density (A/m^2), A_{cell} is the electrolyzer cell area (m^2), and V_{cell} is the electrolyzer cell voltage (V). V_{rev} , $V_{i,activation}$, and V_{ohm} are the electrolyzer reversible, activation, and ohmic overpotential (V), respectively. Also, T_{ez} is the nominal operating electrolyzer temperature (353 K), R is the universal fluid constant ($J/k.mole K$), J_i^{ref} is the current density pre-exponential factor ($1.7 \times 10^5 A/m^2$ for anode, $4.6 \times 10^5 A/m^2$ for cathode), F is Faraday constant (96,485 C/mol), $E_{i,activation}$ is the electrolyzer activation energy (76 kJ/mol for anode - 18 kJ/mol for cathode), σ_{ez} is the electrolyzer ionic conductivity, L is the membrane thickness (μm), and λ_a and λ_c are the water content at the anode and cathode.

The energy required for hydrogen production depends on the higher heating value of hydrogen (HHV_{H_2}) and the electrolyzer efficiency η_{ez} which ranges from 40 % to 66 %. Therefore, the energy required per kg of H_2 production equals $39.4 \frac{kWh}{kg} / \eta_{ez} \%$ [34,35]. PEME is used in this study due to its high efficiency, flexibility in accommodating different loads, and gas purity. Its lifespan ranges from 7 to 10 years. The proposed electrolyzer capacity optimization varies from 50 kW to 400 kW and its specifications are given in Table 2.

3.3.5. Backup generator

This study uses a DG as a backup source to offset renewable energy loss due to weather changes and to assess emissions mitigation potential using a FC as an alternative. The diesel fuel consumption rate (liters/hour) can be calculated based on the amount of energy demand through the following equations [22]:

$$F_{Cr} = f_{Pr} \times P_{gen} \quad (30)$$

$$P_{gen} = k_{gen0} P_{gen,output} + k_{gen1} P_{gen,rated} \quad (31)$$

$$\eta_{gen} \% = \frac{\text{generated electrical energy}}{\text{chemical potential energy}} = \frac{3.6 P_{gen,rated}}{m_{diesel} \times LHV_{diesel}} \quad (32)$$

where f_{Pr} is the diesel price (\$/L). $P_{gen,output}$, and $P_{gen,rated}$ are the generator output, and rated power (kW), respectively. Also, k_{gen0} , and k_{gen1} are the fuel curve intercept coefficient (L/hr/kW rated), and slope (L/hr/kW output), respectively. η_{gen} is the electrical generator efficiency (%), 1 kWh equals 3.6 MJ which is a factor of 3.6, m_{diesel} is the diesel mass flow rate (kg/h), and LHV_{diesel} is the higher heating value of diesel fuel (MJ/kg).

3.3.6. Hydrogen tank

Hydrogen, produced from water electrolysis, is stored in hydrogen tanks that can withstand high gas pressure of 350 bar at ambient temperature for extended periods [36]. The stored hydrogen is used as fuel to power hydrogen FCs. In this study, the capacity optimization of the

proposed hydrogen tank varies from 100 kg to 400 kg and the initial tank level equals 10 % relative to tank size. The HOMER PRO software incorporates compressor specifications as an integral component of the hydrogen storage system.

3.3.7. Hydrogen FC

Hydrogen FCs convert hydrogen's chemical energy into electrical energy through hydrogen decomposition into protons and electrons. The process involves two electrodes and an electrolytic membrane, with protons combining oxygen to produce water and electrons producing electrical charges. FC efficiency equals 60 % over a lifespan of 5–20 years, but they cost five times more to produce 1 kW than DGs. CAT 1600 kVA, 1280 kW hydrogen FC is used. It has no fuel curve intercept, 0.06 kg/h/kW fuel curve slope, and 25 % minimum load ratio.

3.3.8. Electric converter

The converter, consisting of an inverter and rectifier, connects system elements by converting alternating current from the grid, DG, and WTs into direct current to power the electrolyzer. It also works to convert the direct current generated from FCs, PV, electrolyzer, and batteries into alternating current to power the electrical load. The converter power output can be calculated as [29]:

$$P_{inverter} = \eta_{inverter} \times (P_{PV,gen} + P_{ez} + P_{FC} + P_{batt,dch,KiBAM}) \quad (33)$$

$$P_{rectifier} = \eta_{rectifier} \times (P_{WT,gen} + P_{gen} + P_{grid}) \quad (34)$$

where $P_{inverter}$, $P_{rectifier}$ are the inverter, and rectifier output power (kW), respectively. Also, $\eta_{inverter}$, $\eta_{rectifier}$ are the inverter, and rectifier efficiencies (%). P_{FC} is the FC output power (kW), and P_{grid} is the used power from the grid (kW). In this study, the converter capacity optimization is selected by Homer optimizer and its specifications are listed in Table 2.

3.4. Financial details

Net present cost (NPC), levelized cost of energy (LCOE), and levelized cost of hydrogen (LCOH) are important economic factors for comparing different scenarios. According to Homer Pro, NPC is the relation between payments and income over the project lifetime (\$), LCOE is the annual cost of energy provided over the project lifetime (\$) and LCOH is the annual cost of hydrogen production over the project lifetime (\$). The costs can be calculated according to the following equations [22,29]:

$$NPC = \frac{C_{ann,tot}}{CRF(i, t)} \quad (35)$$

$$NPV = \frac{NCF}{(1+i)^t} - IC \quad (36)$$

$$CRF(i, t) = \frac{i(1+i)^t}{(1+i)^t - 1} \quad (37)$$

$$i = \frac{i' - f}{1 + f} \quad (38)$$

$$LCOE = \frac{\sum_{t=0}^t \frac{IC+OM+RP}{(1+i)^{tC}}}{\sum_{t=1}^t \frac{E_{ann,t}}{(1+d)^{tC-1}}} \quad (39)$$

$$LCOH = \frac{C_{electrolysis} + C_{ez,ip}}{\sum_{t=1}^{t_{ez}} H_{ez}} \quad (40)$$

$$C_{electrolysis} = LCOE_h \times \frac{\sum_{t=1}^t E_{ann,t}}{t} \quad (41)$$

$$C_{ez,ip} = C_{ez} \times \frac{H_{ez} + P_{ez}}{t \times CF \times \eta_{ez} \times 8760} \quad (42)$$

where $C_{ann,tot}$ is the total project cost per year (\$/year), $CRF(i, t)$ is the capital recovery factor over the project lifetime, i is the real interest rate (%), $\tilde{i}$ is the nominal interest rate (%), f is the annual inflation rate, t is the project lifetime (year), and d is the components degradation factor. Net present value (NPV) represents the cost of revenue or net cash flow (NCF) subtracted from initial cost (IC). In addition, OM, and RP are the annual operation, maintenance, and replacement costs (\$). $E_{ann,1}$, and $E_{ann,n}$ represent the electrical energy obtained after one, and n th years (kWh/year). For LCOH calculation, $C_{electrolysis}$ is the cost of the electrolysis process (\$), $C_{ez,ip}$ is the cost of electrolyzer input power (\$), H_{ez} is the hydrogen production (kg), t_{ez} is the electrolyzer lifetime (year), $LCOE_H$ is the levelized cost of electricity used for hydrogen production (\$), C_{ez} is the cost of electrolyzer (\$/unit), and CF is electrolyzer capacity factor (%). According to Ref. [37], Homer's operating and technical parameters of this study are listed in Table 3. Also, Table 4 shows the financial details of each component.

In this study, water is considered the primary medium for hydrogen production via electrolysis. As seawater is the most accessible source, it must first undergo desalination. The HOMER software assumes that the electrolyzer input is already purified, thereby excluding desalination costs from the optimization process. To address this limitation, a supplementary analysis was conducted to estimate the cost of seawater desalination for hydrogen production, as expressed by the following equations [18]:

$$C_{des} = C_{H_2O} \times Q_{H_2O} \times Q_{H_2} \quad (43)$$

$$C_{des,T} = C_{des} \times t \quad (44)$$

$$C_{Pr} = \frac{C_{des,T}}{NPC + C_{des,T}} \quad (45)$$

where, C_{des} is the annual cost of water desalination (\$/year), C_{H_2O} is the unit cost of desalinated water, taken as 1 \$/m³ according to Kettani et al. [39], Q_{H_2O} is the water consumption for producing 1 kg of hydrogen (0.009 m³/kg), and Q_{H_2} is the annual hydrogen production (kg/year). $C_{des,T}$ denotes the total desalination cost over the project lifetime (\$), and C_{Pr} is the cost fraction, used to assess the relative contribution of water desalination to the overall NPC. It should be noted that the cost of water desalination is negligible, accounting for only 0.13–1.11 % of the NPC.

In this study, data quality and availability, along with valuation errors, may represent significant sources of error. The quality and availability of data, encompassing variables such as the market risk premium, debt ratio, and tax rate, can fluctuate depending on the data source, time period, industry, and geographic region. The issue requires the use of diverse data sources and regular updates of this information. Valuation errors arise from the challenge of accurately assigning values to critical project elements like environmental quality and climate change impacts. These errors can be mitigated by applying appropriate methodologies, including contingent valuation and shadow pricing.

4. Results and discussion

The study explores the establishment of a plant in four coastal cities for renewable energy production, using electricity for loads and hydrogen for arid areas using FCs as shown in Fig. 7. The study explores four hydrogen production scenarios, including solar panels, water tanks, storage batteries, proton exchange membrane electrolyzers, hydrogen tanks, and DGs as standalone and grid-connected hybrid systems. In each scenario, the effect of different WT numbers on energy production is studied and three categories are used: 15, 20, and 50 WTs. Homer Pro is used to simulate and optimize the system. According to Homer's calculation report, 176,476 solutions were simulated, 3268 were

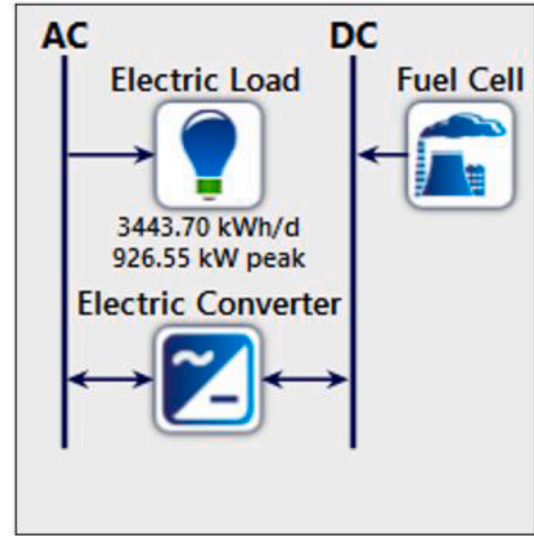


Fig. 7. Hydrogen stored as a primary fuel source.

feasible and the rest were infeasible due to the capacity shortage constraint and maximum unmet hydrogen. The comparison of system costs, electricity, hydrogen production, electrolyzer power, load served, shortage percentage, hydrogen tank filling levels, excess energy, and renewable fraction is conducted. Fig. 8 shows the different scenario configurations.

4.1. Scenario 1: stand-alone PV/WT/battery/PEME/H₂ tank hybrid system

The annual energy analysis of the proposed system for each site study is presented in Table 5. Photovoltaic cells and WTs are used to supply the required energy. Batteries are also used to enhance the system's stability by providing energy during weather fluctuations and at night. The electrolyzer uses the generated energy to produce hydrogen and then stores it in tanks in preparation for transporting it to be reused in producing electrical energy in places where there are no permanent water sources. The specific consumption of 1 kg of hydrogen is 46.4 kWh. This system is studied with three cases 15, 20, and 50 WTs.

In New Alamein (case 1: 15 WTs), PV produces 37.3 % of the energy while the WTs produce 62.7 % of the energy. The energy is used at a rate of 34 % to supply electrical loads and 66 % to operate the electrolyzer and produce hydrogen. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$44.62 million, 0.3145 \$/kWh, and 8.88 \$/kg, respectively. The system suffers from a capacity shortage of 0.0710 %. In case 2 (20 WTs), PV produces 14.9 % of the energy while the WTs produce 85.1 % of the energy. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$24.94 million, 0.1758 \$/kWh, and 4.96 \$/kg, respectively. The system suffers from a capacity shortage of 0.0852 %. In case 3 (50 WTs), PV produces 2.51 % of the energy while the WTs produce 97.5 % of the energy. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$8.46 million, 0.0557 \$/kWh, and 1.68 \$/kg, respectively. The system suffers from a capacity shortage of 0.0997 %.

In El Tor, no feasible solutions were found for case 1 due to not enough generation capacity. In case 2, PV produces 39.3 % of the energy while the WTs produce 60.7 % of the energy. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$50.47 million, 0.3556 \$/kWh, and 10 \$/kg, respectively. The system suffers from a capacity shortage of 0.0154 %. In case 3, PV produces 7.39 % of the energy while the WTs produce 92.6 % of the energy. The system net present cost, the levelized cost of energy, and the levelized

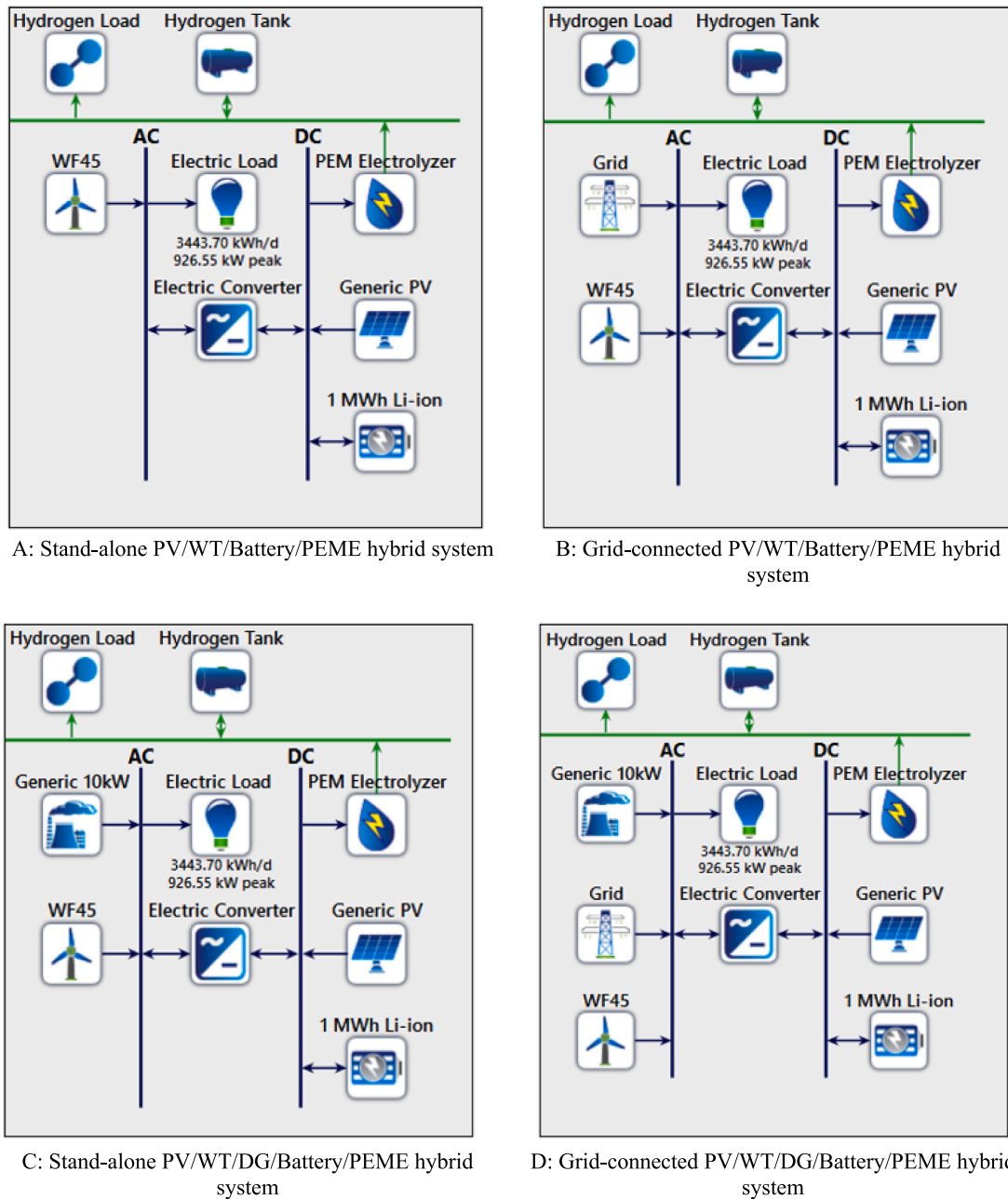


Fig. 8. Proposed configurations for all sites studied.

cost of hydrogen are \$14.85 million, 0.0115 \$/kWh, and 2.95 \$/kg, respectively. The system suffers from a capacity shortage of 0.0514 %.

In Salam (case 1), PV produces 48.1 % of the energy while WTs produce 51.9 % of the energy. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$37.89 million, 0.2671 \$/kWh, and 7.53 \$/kg, respectively. The system suffers from a capacity shortage of 0.0945 %. In case 2, PV produces 26.8 % of the energy while the WTs produce 73.2 % of the energy. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$34.11 million, 0.2404 \$/kWh, and 6.78 \$/kg, respectively. The system has no capacity shortage. In case 3, PV produces 4.46 % of the energy while the WTs produce 95.5 % of the energy. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$12.20 million, 0.08597 \$/kWh, and 2.43 \$/kg, respectively. The system has no capacity shortage.

In Suez (case 1), PV produces 41.2 % of the energy while WTs

produce 58.8 % of the energy. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$45.93 million, 0.3238 \$/kWh, and 9.14 \$/kg, respectively. The system suffers from a capacity shortage of 0.0819 %. In case 2, PV produces 21.1 % of the energy while the WTs produce 78.9 % of the energy. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$29.12 million, 0.2052 \$/kWh, and 5.79 \$/kg, respectively. The system suffers from a capacity shortage of 0.0938 %. In case 3, PV produces 4.37 % of the energy while the WTs produce 95.6 % of the energy. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$10.999 million, 0.07754 \$/kWh, and 2.19 \$/kg, respectively. The system suffers from a capacity shortage of 0.0996 %.

Fig. 9 illustrates the share of energy from renewables in all cases for each site study. The results show that WTs contribute the largest portion of energy production. However, as the number of WTs increases, the

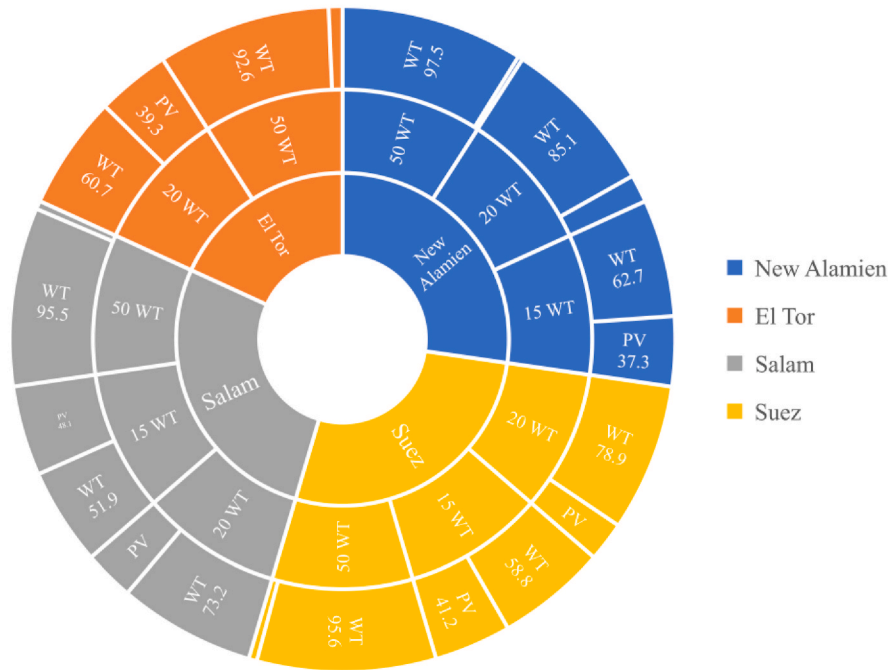


Fig. 9. Proportions of energy from REs in a stand-alone PV/WT/Battery/PEME/H₂ tank hybrid system in all sites studied.

capacity shortage increases due to the different weather fluctuations in wind speed throughout the year. Additionally, battery systems play a crucial role in mitigating these shortages by ensuring a stable energy supply throughout the year. It can be understood from these results that case 3 has the lowest cost in terms of net present cost, levelized cost of energy, and levelized cost of hydrogen.

4.2. Scenario 2: grid-connected PV/WT/battery/PEME/H₂ tank hybrid system

The proposed system can be connected to the grid to sell the excess electricity. The grid is used as an infinite capacity battery to enhance the system cost. Its revenue reduces the system NPC, LCOE, and LCOH. The grid electricity price equals 0.10 \$/kWh, and its sellback price equals 0.05 \$/kWh.

In New Alamein (case 1), 25.4 % of the energy is used to supply electrical loads, 49.4 % to operate the electrolyzer, and 25.2 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$37.59 million, 0.1331 \$/kWh, and 7.48 \$/kg, respectively. In case 2, 25.4 % of the energy is used to supply electrical loads, 49.5 % to operate the electrolyzer, and 25.1 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$17.93 million, 0.06359 \$/kWh, and 3.57 \$/kg, respectively. In case 3, 25.2 % of the energy is used to supply electrical loads, 49 % to operate the electrolyzer, and 25.8 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$1.14 million, 0.003983 \$/kWh, and 0.227 \$/kg, respectively.

In El Tor, no feasible solutions were found for case 1 due to not enough generation capacity. In case 2, 27.3 % of the energy is used to supply electrical loads, 45.5 % to operate the electrolyzer, and 27.2 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$43.42 million, 0.1534 \$/kWh, and 8.62 \$/kg, respectively. In case 3, 27 % of the energy is used to supply electrical loads, 45.1 % to operate the electrolyzer, and 27.9 % for sale to the grid. The system net present cost, the levelized cost of energy and the levelized cost of hydrogen are \$7.51 million, 0.02603 \$/kWh, and 1.49 \$/kg, respectively.

In Salam (case 1), 26.5 % of the energy is used to supply electrical loads, 48.2 % to operate the electrolyzer, and 25.3 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$31.14 million, 0.1125 \$/kWh, and 6.19 \$/kg, respectively. In case 2, 26.5 % of the energy is used to supply electrical loads, 48.1 % to operate the electrolyzer, and 25.4 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$27.31 million, 0.09820 \$/kWh, and 5.43 \$/kg, respectively. In case 3, 26.3 % of the energy is used to supply electrical loads, 47.7 % to operate the electrolyzer, and 26 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$5.19 million, 0.01841 \$/kWh, and 1.03 \$/kg, respectively.

In Suez (case 1), 26.8 % of the energy is used to supply electrical loads, 47.4 % to operate the electrolyzer, and 25.8 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$39.16 million, 0.1405 \$/kWh, and 7.79 \$/kg, respectively. In case 2, 26.7 % of the energy is used to supply electrical loads, 47.3 % to operate the electrolyzer, and 26 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$22.20 million, 0.07928 \$/kWh, and 4.42 \$/kg, respectively. In case 3, 26.4 % of the energy is used to supply electrical loads, 46.8 % to operate the electrolyzer, and 26.8 % for sale to the grid. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$3.79 million, 0.01327 \$/kWh, and 0.754 \$/kg, respectively.

Fig. 10 shows the share of energy from renewables and grid sell-back in all cases for each site study. The results show that revenue from grid sell-back reduces the system cost by about 65 % compared to Scenario 1 (Stand-alone PV/WT/Battery/PEME/H₂ tank hybrid system). It is evident from these results that case 3 has the lowest cost in terms of net present cost, levelized cost of energy, and levelized cost of hydrogen.

4.3. Scenario 3: stand-alone PV/WT/DG/battery/PEME/H₂ tank hybrid system

This scenario relies on adding a DG to be used as a backup source to compensate for the loss of power from renewable sources during times of

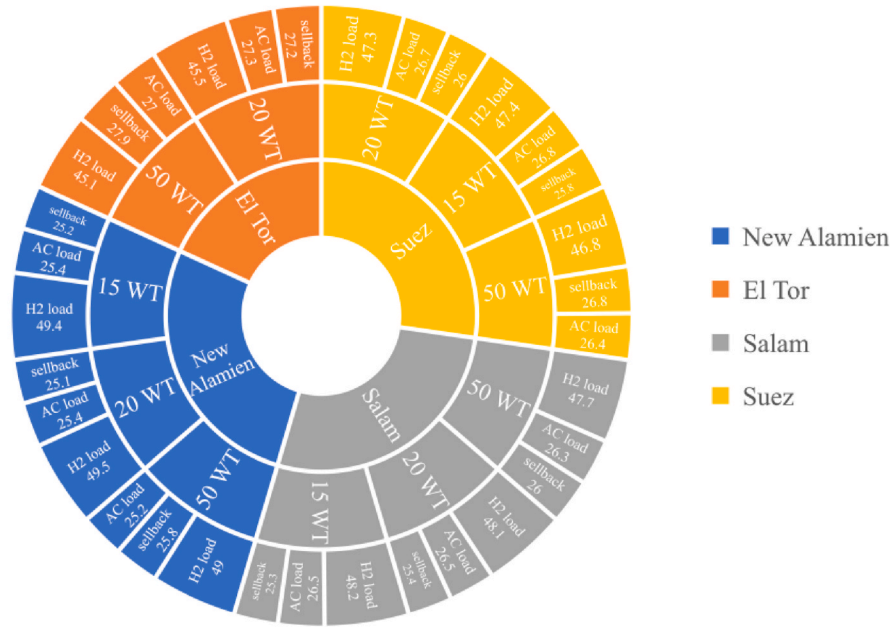


Fig. 10. Share of energy in a grid-connected PV/WT/Battery/PEME/H₂ tank system in all sites.

weather fluctuations. Unfortunately, DG exhaust contains many toxic gases such as carbon monoxide (CO), carbon dioxide (CO₂), unburned hydrocarbons (UHC_s), particulate matter (PM), sulfur dioxide (SO₂), and nitrogen oxides (NO). Producing 1 kWh produces 632 g of CO₂ and 2.74 g of NO [40]. Therefore, RESs are relied upon to produce hydrogen, with the deficit being filled by a DG. Emission rates varied from one city to another, as the first case achieved the lowest emission rate in New Alamein, the second case achieved the lowest emission rate in El Tor, and the third case achieved the lowest emission rate in New Port Said. Table 6 shows the gas emissions in all cases for each site under study. Therefore, the optimal case for hydrogen production is determined based on the amount of energy produced at the lowest cost. The annual energy analysis of the proposed system for each site study is presented in Table 7. The results illustrate that case 3 has the lowest cost in terms of net present cost, levelized cost of energy, and levelized cost of hydrogen.

4.4. Scenario 4: grid-connected PV/WT/DG/battery/PEME/H₂ tank hybrid system

The proposed system in scenario 3 generates a large amount of surplus electricity. Connecting the system to the grid results in selling the surplus electricity to reduce the system cost and minimize system losses.

In New Alamein, case 1 achieves an excess electricity ratio of 85.5 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$37.54 million, 0.1329 \$/kWh, and 7.47 \$/kg, respectively. Case 2 achieves an excess electricity ratio of 85.2 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$17.88 million, 0.06341 \$/kWh, and 3.56 \$/kg, respectively. Case 3 achieves an excess electricity ratio of 93.1 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$0.89 million, 0.003104 \$/kWh, and 0.177 \$/kg, respectively.

In El Tor, no feasible solutions were found for case 1 due to not enough generation capacity. Case 2 achieves an excess electricity ratio of 88.7 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$43.37 million, 0.1532 \$/kWh, and 8.61 \$/kg, respectively. Case 3 achieves an excess electricity ratio of 93 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$7.46 million, 0.02585 \$/kWh, and 1.48 \$/kg, respectively.

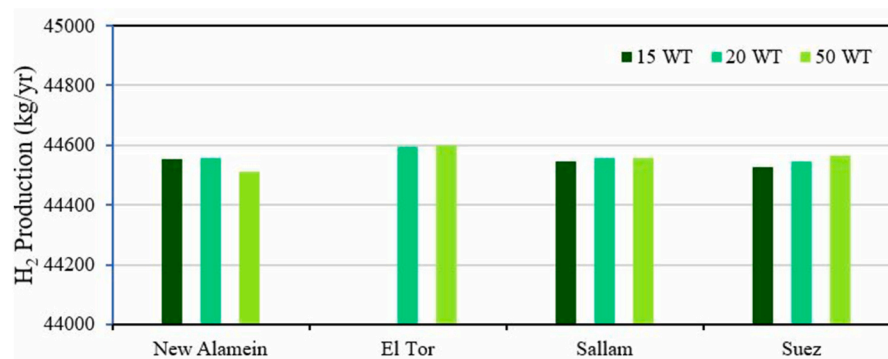
In Salam, case 1 achieves an excess electricity ratio of 86.1 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$31.09 million, 0.1123 \$/kWh, and 6.18 \$/kg, respectively. Case 2 achieves an excess electricity ratio of 85.7 %. The

Table 6
Comparison of generator performance and gas emissions in all cases for each site studied.

WT	PV/WT/DG/Battery/PEME/H ₂ tank									
	DG performance				Emissions (kg/yr)					
	Operation (hrs/yr)		Starts (start/yr)	Fuel consumption (L)	CO ₂	CO	UHC _s	PM	SO ₂	NO
New Alamein	15	10	6	22.7	59.2	0.448	0.0163	0.0272	0.145	0.509
	20	12	7	29.4	76.7	0.580	0.0211	0.0352	0.188	0.659
	50	13	7	33.4	87.4	0.661	0.0241	0.0401	0.214	0.751
El Tor	15	–	–	–	–	–	–	–	–	–
	20	3	3	5.73	15	0.113	0.00413	0.00069	0.0367	0.129
	50	7	5	14.8	38.7	0.292	0.0107	0.0177	0.0949	0.332
Salam	15	12	7	29.4	76.7	0.580	0.0211	0.0352	0.188	0.659
	20	14	8	33.9	88.5	0.670	0.0244	0.0406	0.217	0.761
	50	0	0	0	0	0	0	0	0	0
Suez	15	13	6	30.6	79.8	0.604	0.0220	0.0366	0.196	0.686
	20	14	6	32.1	83.8	0.634	0.0231	0.0384	0.206	0.721
	50	19	9	48.4	127	0.957	0.0349	0.0580	0.311	1.09

Table 7Energy analysis of a stand-alone PV/WT/DG/Battery/PEME/H₂ tank system for each site studied.

WT		PV/WT/DG/Battery/PEME/H ₂ tank						Hydrogen (kg/yr)		Autonomy (hr.)	
		Energy (kWh/yr)									
		PV	WT	DG	Battery	AC load	Shortage	PEME	H ₂ load	H ₂ tank	Battery
New Alamein	15	13,031,123	21,883,028	62.5	340,253	1,256,502	790	44,524	44,286	92.9	22.3
	20	5,031,769	29,177,371	82.5	334,833	1,256,466	954	44,533	44,285	92.9	22.3
	50	1,745,962	72,943,426	95.1	318,687	1,256,314	1143	44,551	44,285	92.9	22.3
El Tor	15	–	–	–	–	–	–	–	–	–	–
	20	16,390,562	25,272,174	15	346,667	1,256,897	163	44,593	44,285	92.9	22.3
	50	5,044,255	63,180,435	40	325,239	1,256,836	576	44,598	44,285	92.9	22.3
Salam	15	16,825,337	18,185,530	82.5	387,286	1,256,356	1068	44,542	44,285	92.9	22.3
	20	9,857,501	24,247,373	95	379,428	1,256,378	1257	44,595	44,328	92.9	22.3
	50	2,830,222	60,618,433	0	360,623	1,256,950	0	44,556	44,285	92.9	27.9
Suez	15	13,830,210	19,737,120	85	365,673	1,256,367	902	44,507	44,285	92.9	22.3
	20	7,035,474	26,316,159	88.7	358,187	1,256,373	1052	44,524	44,285	92.9	22.3
	50	2,864,284	65,790,398	138	335,591	1,256,258	1196	44,552	44,285	92.9	22.3

**Fig. 11.** Annual green hydrogen production of grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid system in all cases for each site studied. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)**Table 8**

Cost comparison for different scenarios at New Alamein City.

Cost type	Value	PV/WT/Battery/PEME/H ₂ tank						PV/WT/DG/Battery/PEME/H ₂ tank					
		Stand-alone			Grid-connected			Stand-alone			Grid-connected		
	WT	15	20	50	15	20	50	15	20	50	15	20	50
NPC	\$M	44.6	24.9	8.46	37.6	17.9	1.14	44.6	24.9	8.11	37.5	17.9	0.89
IC	\$M	30.5	20.5	30.2	30.5	20.5	30.2	30.5	20.4	30	30.5	20.5	30
LCOE	\$/kWh	0.315	0.176	0.0596	0.133	0.0636	0.0039	0.314	0.176	0.0572	0.133	0.0634	0.00310
LCOH	\$/kg	8.88	4.96	1.68	7.48	3.57	0.227	8.87	4.95	1.61	7.47	3.56	0.177
AES	\$K	–	–	–	62.25	62.08	64.17	–	–	–	62.25	62.08	63.96
DS	–	CC	CC	CC	CC	CC	CC	LF	LF	LF	LF	LF	LF

CC: Cycle charging, IC: Initial capital cost, AES: Annual energy sold to grid, DS: Dispatch strategy.

system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$27.59 million, 0.09905 \$/kWh, and 5.48 \$/kg, respectively. Case 3 achieves an excess electricity ratio of 92.2 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are -\$5.14 million, 0.01822 \$/kWh, and 1.02 \$/kg, respectively.

In Suez, case 1 achieves an excess electricity ratio of 85.7 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$39.11 million, 0.1403 \$/kWh, and 7.78 \$/kg, respectively. Case 2 achieves an excess electricity ratio of 85.5 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are \$22.15 million, 0.07910 \$/kWh, and 4.41 \$/kg, respectively. Case 3 achieves an excess electricity ratio of 92.9 %. The system net present cost, the levelized cost of energy, and the levelized cost of hydrogen are -\$3.66 million, 0.01283 \$/kWh, and 0.728 \$/kg, respectively.

For the grid-connected hybrid system consisting of PV/WT/DG/Battery/PEME/H₂ tank, Fig. 11 illustrates the annual green hydrogen production in all cases for each site studied. It can be understood from these results that case 3 has the lowest cost due to the revenue generated from selling the surplus renewable energy to the utility grid.

For comparing the cost types for different scenarios across all case studies, scenario 4 (grid-connected PV/50 WT/DG/Battery/PEME/H₂ tank) achieves the lowest cost making it a viable solution. The cost comparison for different scenarios in all cases is shown in Tables 8–11.

For direct cross-comparison across the studied locations and configurations, Fig. 12 has been included. This figure offers a visual representation that identifies the site and system configuration yielding the most cost-effective performance.

This study investigates several energy system configurations by analyzing various operational scenarios. Initially, a system is developed based on water electrolysis, utilizing solar and wind energy sources. The

Table 9

Cost comparison for different scenarios at El Tor City.

Cost type	Value	PV/WT/Battery/PEME/H ₂ tank						PV/WT/DG/Battery/PEME/H ₂ tank					
		Stand-alone			Grid-connected			Stand-alone			Grid-connected		
	WT	15	20	50	15	20	50	15	20	50	15	20	50
NPC	\$M	–	50.5	14.8	–	43.4	7.51	–	50.4	14.8	–	43.4	7.46
IC	\$M	–	37.5	34.9	–	37.5	34.9	–	37.5	34.9	–	37.5	34.9
LCOE	\$/kWh	–	0.356	0.105	–	0.153	0.0260	–	0.355	0.104	–	0.153	0.0259
LCOH	\$/kg	–	10.0	2.95	–	8.62	1.49	–	10.0	2.94	–	8.61	1.48
AES	\$K	–	–	–	–	62.47	64.95	–	–	–	–	62.47	64.95
DS	–	–	CC	CC	–	CC	CC	–	LF	LF	–	LF	LF

Table 10

Cost comparison for different scenarios at New Port Said “Salam” City.

Cost type	Value	PV/WT/Battery/PEME/H ₂ tank						PV/WT/DG/Battery/PEME/H ₂ tank					
		Stand-alone			Grid-connected			Stand-alone			Grid-connected		
	WT	15	20	50	15	20	50	15	20	50	15	20	50
NPC	\$M	37.9	34.1	12.2	31.1	27.3	5.19	37.8	34.4	12.2	31.1	27.6	5.14
IC	\$M	35.2	25.7	31.6	35.2	25.7	31.6	35.2	27.0	31.6	35.2	27.0	31.6
LCOE	\$/kWh	0.267	0.240	0.0860	0.112	0.0982	0.0184	0.267	0.243	0.0856	0.112	0.0990	0.0182
LCOH	\$/kg	7.53	6.78	2.43	6.19	5.43	1.03	7.52	6.84	2.42	6.18	5.48	1.02
AES	\$K	–	–	–	59.79	60.29	62.08	–	–	–	59.79	60.52	62.08
DS	–	CC	CC	CC	CC	CC	CC	LF	LF	CC	LF	LF	CC

IC: Initial capital cost, AES: Annual energy sold to grid, DS: Dispatch strategy.

Table 11

Cost comparison for different scenarios at Suez City.

Cost type	Value	PV/WT/Battery/PEME/H ₂ tank						PV/WT/DG/Battery/PEME/H ₂ tank					
		Stand-alone			Grid-connected			Stand-alone			Grid-connected		
	WT	15	20	50	15	20	50	15	20	50	15	20	50
NPC	\$M	45.9	29.1	11.0	39.2	22.2	3.79	45.9	29.1	10.9	39.1	22.2	3.66
IC	\$M	31.0	23.1	31.9	31.1	23.1	31.9	31.1	23.1	31.7	31.1	23.1	31.7
LCOE	\$/kWh	0.324	0.205	0.0775	0.141	0.0793	0.0133	0.323	0.205	0.0765	0.140	0.0791	0.0128
LCOH	\$/kg	9.14	5.79	2.19	7.79	4.42	0.754	9.13	5.78	2.16	7.78	4.41	0.728
AES	\$K	–	–	–	60.60	61.21	63.81	–	–	–	60.60	61.21	63.66
DS	–	CC	CC	CC	CC	CC	CC	LF	LF	LF	LF	LF	LF

system is coupled with battery storage to retain surplus electricity, ensuring power availability during nighttime hours and periods of weather-induced fluctuations. Additionally, the hydrogen produced is stored in dedicated tanks for transportation to areas remote from water sources, enabling its use in energy generation.

The analysis reveals that increasing the number of WTs leads rise in electricity shortages due to variability in weather conditions. Nevertheless, the proposed system demonstrates substantial cost savings, achieving reductions in the net present cost, levelized cost of energy, and levelized cost of hydrogen by 81 % in New Alamein, 71 % in El Tor, 68 % in New Port Said, and 76 % in Suez.

To mitigate electricity shortages during nighttime and adverse weather conditions, the system is augmented with a DG operating as a backup power source. This modification results in a decrease in power shortages, ranging from 9 to 12 % in New Alamein, 11–16 % in El Tor, 10–100 % in New Port Said, and 5–13 % in Suez.

Further improvements are made by integrating the system with the national electricity grid, allowing for the sale of surplus electricity and further cost optimization. As a result, the net present cost and levelized cost of hydrogen are reduced by up to 89.5 %, while the levelized cost of energy declines by 94.8 % in New Alamein. Corresponding reductions in El Tor are 50 % and 75 %; in New Port Said, 58 % and 79 %; and in Suez, 67 % and 84 %, respectively.

It can be understood from these results that the third case of scenario 4 is the winning case. It achieves the best results in hydrogen production based on the lowest NPC, LCOE, and LCOH.

4.5. Winning case analysis

Through this section, the results of the optimal case used for hydrogen production will be discussed. The results are analyzed through system architecture, energy analysis, hydrogen analysis, excess electricity, and economic evaluation.

4.5.1. System overview

The proposed hybrid system is based on renewable energies with batteries, an electrolyzer, a hydrogen tank, and a generator as backup source. The system architecture and overall production are shown in Table 12. The system production mainly relies on WTs in Mediterranean cities to increase wind activity throughout the year, while the reliance on solar cells increases in Red Sea cities to increase solar radiation.

4.5.2. Energy analysis

Energy analysis is based on the component capacity factor, operating hours, electrical efficiency and autonomy as energy is not purchased from the grid. Capacity factor is the ratio of energy produced to the maximum energy during a certain time (%). Electrical efficiency is the ratio of the electrical energy to the fuel energy during a certain time (%). Autonomy is the relationship between the maximum energy capacity and the electric load (hr). Table 13 shows the monthly energy production of the grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid system for each site studied.

In New Alamein, the electrical energy obtained from the system is the largest in Fall/Winter compared to Spring/Summer due to the high

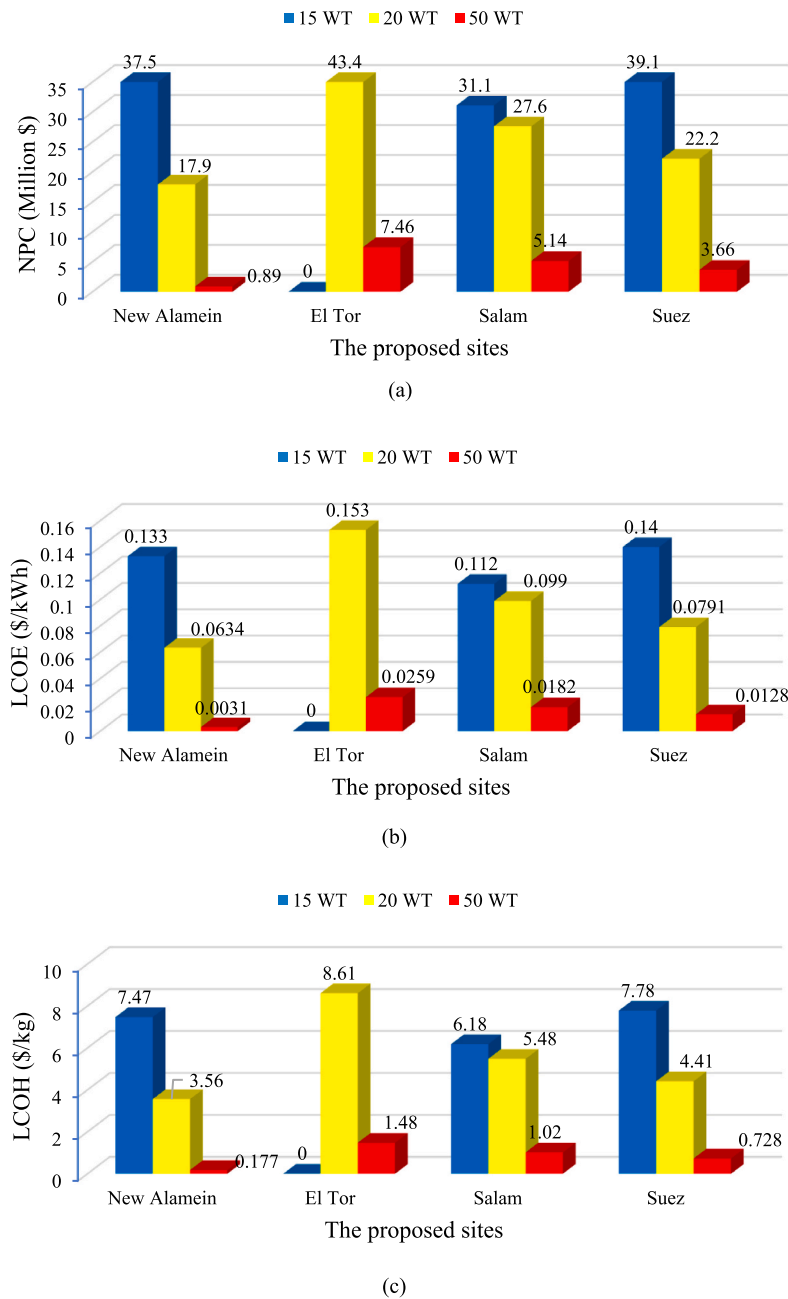


Fig. 12. Comparison of the grid-connected PV/WT/DG/Battery/PEME/H₂-tank hybrid system across the studied locations and system configurations with respect to: (a) NPC, (b) LCOE, and (c) LCOH.

Table 12

Optimal configuration for each site studied with 400 kW electrolyzer capacity at 50 WT.

City	Architecture						Production		
	PV (kW)	WT (kW)	DG (kW)	Battery (kWh)	H ₂ tank capacity (kg)	Converter (kW)	Electricity (kWh/yr)	Hydrogen (kg/yr)	RF (%)
New Alamein	1107	2,5000	10	4	400	1025	74,686,949	44,551	100
El Tor	3082	2,5000	10	4	400	847	68,224,730	44,598	100
Salam	1686	2,5000	10	5	400	1050	63,448,654	44,556	100
Suez	1748	2,5000	10	4	400	1100	68,654,820	44,552	100

wind speed activity. The annual energy production is about 74.7 GWh/yr. The capacity factor of solar panels, WTs, generator, and converter are 18 %, 33.3 %, 0.109 %, and 18.5 %, respectively. Also, the operating hours of solar panels, WTs, and generator are 4384 h/yr, 6762 h/yr, and

13 h/yr. The generator's mean electrical efficiency is 28.9 %. Battery autonomy is 22.3 h.

In El Tor, the electrical energy obtained from the system is the largest in Spring/Summer compared to Fall/Winter due to the large amount of

Table 13

Monthly energy production (MWh) of the grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid system for each site studied.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
New Alamein												
PV	117.5	123.1	152.2	162.3	164.1	160.9	167.5	167.7	155.9	141.8	117.2	113.3
WT	6969.6	6823.1	6914.5	6364.7	5793.4	5926.6	6419	5470.8	5241.6	5257.9	5238.9	6523.3
DG	0	0.029	0.02	0.003	0	0.003	0	0.029	0	0.013	0	0
El Tor												
PV	386.8	388.6	466	450.96	457.3	448.7	454.9	453.5	430.4	401.9	351.1	354.01
WT	4161.7	4531.3	5394.4	5071.6	5960.2	7378.6	6794.4	6876.34	6060.4	3942.6	3213.6	3795.2
DG	0.01	0	0	0	0.0025	0	0.0125	0	0	0.0125	0	0.0025
Salam												
PV	192.02	193.2	251.1	257.4	266.8	259.1	262.5	265.7	257.6	242.3	200.8	181.5
WT	6309.2	6086.7	6515.2	5567.1	4601.3	4283.9	4214.6	3608.2	3834.6	4468.4	5117.3	6011.8
DG	0	0	0	0	0	0	0	0	0	0	0	0
Suez												
PV	195.5	203.1	265.1	263.1	266.3	260.6	265.6	266.7	254.2	236.99	199.5	187.3
WT	4740.8	4768.4	5950.4	5700.7	6058.1	6376.1	5302.97	5379.3	6202.7	5737.1	4758.2	4815.6
DG	0.0325	0.0525	0.02	0.0025	0	0.0025	0	0.015	0	0.0125	0	0

solar radiation as well as the high wind speed activity. The annual energy production is about 68.3 GWh/yr. The capacity factor of solar panels, WTs, generator, and converter are 18.7 %, 28.8 %, 0.0457 %, and 16.9 %, respectively. Also, the operating hours of solar panels, WTs, and generator are 4387 h/yr, 6592 h/yr, and 7 h/yr. The generator's mean electrical efficiency is 27.5 %. Battery autonomy is 22.3 h.

In Salam, the electrical energy obtained from the system is the least in Spring/Summer compared to Fall/Winter due to the lack of solar radiation and wind speed. The annual energy production is about 63.4 GWh/yr. The capacity factor of solar panels, WTs, converter, and electrolyzer are 19.2 %, 27.7 %, 15.2 %, and 65.1 %, respectively. Also, the operating hours of solar panels, and WTs are 4387 h/yr, and 6397 h/yr. Battery autonomy is 27.9 h.

In Suez, the electrical energy obtained from the system is stable throughout the year with an annual energy production of about 68.7 GWh/yr. The capacity factor of solar panels, WTs, generator, converter, and electrolyzer are 18.7 %, 30 %, 0.157 %, 13.9 % and 63.5 %, respectively. Also, the operating hours of solar panels, WTs, and generator are 4386 h/yr, 6580 h/yr, and 19 h/yr. The generator's mean electrical efficiency is 28.8 %. Battery autonomy is 22.3 h. It can be understood from these results that WT has the largest share of energy and generator has a limit to share energy which reduces gas emissions.

4.5.3. Hydrogen analysis

Hydrogen production is based on the electrolyzer capacity factor, operating hours, annual input energy, and hydrogen tank autonomy. The annual energy sent to the electrolyzer is 2.1 GWh/yr. The hydrogen tank level at the beginning of the year is 40 kg. Fig. 13 shows the monthly profile of the hydrogen level tank for grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid system for the site studied. The red color represents the high energy production by the hybrid system due to high solar radiation availability and wind speed activity. Blue and black indicate low energy production due to fog and bad weather.

In New Alamein, the hydrogen production quantity is the largest in Fall/Winter compared to Spring/Summer. The annual hydrogen production is about 44.551 kg/yr. The capacity factor and operating hours of the electrolyzer are 69.7 %, and 6605 h/yr, respectively. The hydrogen tank content at the end of the year and autonomy are 306 kg, and 92.9 h, respectively.

In El Tor, the hydrogen production quantity is the largest in Spring/Summer compared to Fall/Winter. The annual hydrogen production is about 44.598 kg/yr. The capacity factor and operating hours of the electrolyzer are 59.8 %, and 6620 h/yr, respectively. The hydrogen tank content at the end of the year and autonomy are 353 kg, and 92.9 h, respectively.

In Salam, the hydrogen production quantity is the least in Spring/Summer compared to Fall/Winter. The annual hydrogen production is about 44.556 kg/yr. The capacity factor and operating hours of the electrolyzer are 65.1 %, and 6390 h/yr, respectively. The hydrogen tank content at the end of the year and autonomy are 311 kg, and 92.9 h, respectively.

In Suez, the hydrogen production quantity is stable throughout the year with an annual hydrogen production of about 44.552 kg/yr. The capacity factor and operating hours of the electrolyzer are 63.5 %, and 6558 h/yr, respectively. The hydrogen tank content at the end of the year and autonomy are 307 kg, and 92.9 h, respectively. Obviously, the results indicate that the hydrogen tank level is full throughout the year in all sites studied.

4.5.4. Excess electricity

The system's net present cost can be reduced by selling excess electricity to the grid. Fig. 14 shows the monthly energy sold to the grid for grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid system for each site studied. The annual energy sold to the grid is about 1.27 GWh/yr. Most of the energy sold to the grid is during the sunny daytime hours from 6 a.m. to 7 p.m. due to the availability of solar radiation and active

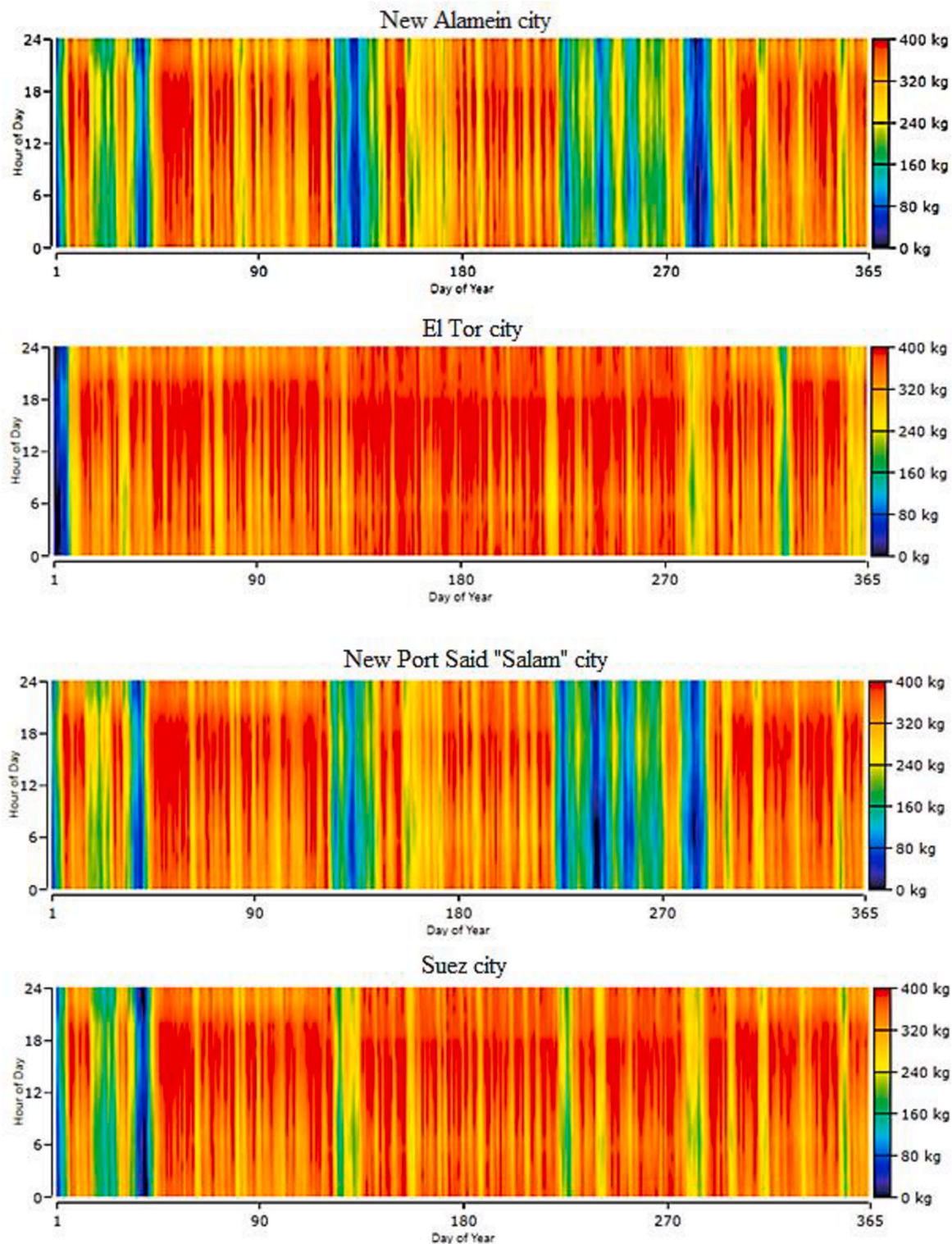


Fig. 13. Monthly profile of the quantity of hydrogen in the storage tank for grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid system for the optimal case of all sites studied.

wind speed. Also, high wind speeds at night lead to some of the energy being sold to the grid.

4.5.5. Economic evaluation

Economic evaluation is based on the cost of the project cost over its lifetime. Capital, operation, replacement, salvage, and fuel costs are used to evaluate the financial details of the project. System components

such as solar panels and hydrogen tanks are durable. On the other hand, WTs, electrolyzers, and batteries need to be replaced multiple times over the project's lifetime. The salvage cost can be used to offset the replacement component cost. Fig. 15 illustrates the cost types of grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid systems for each site studied.

Beyond the financial aspects of system components, it is also

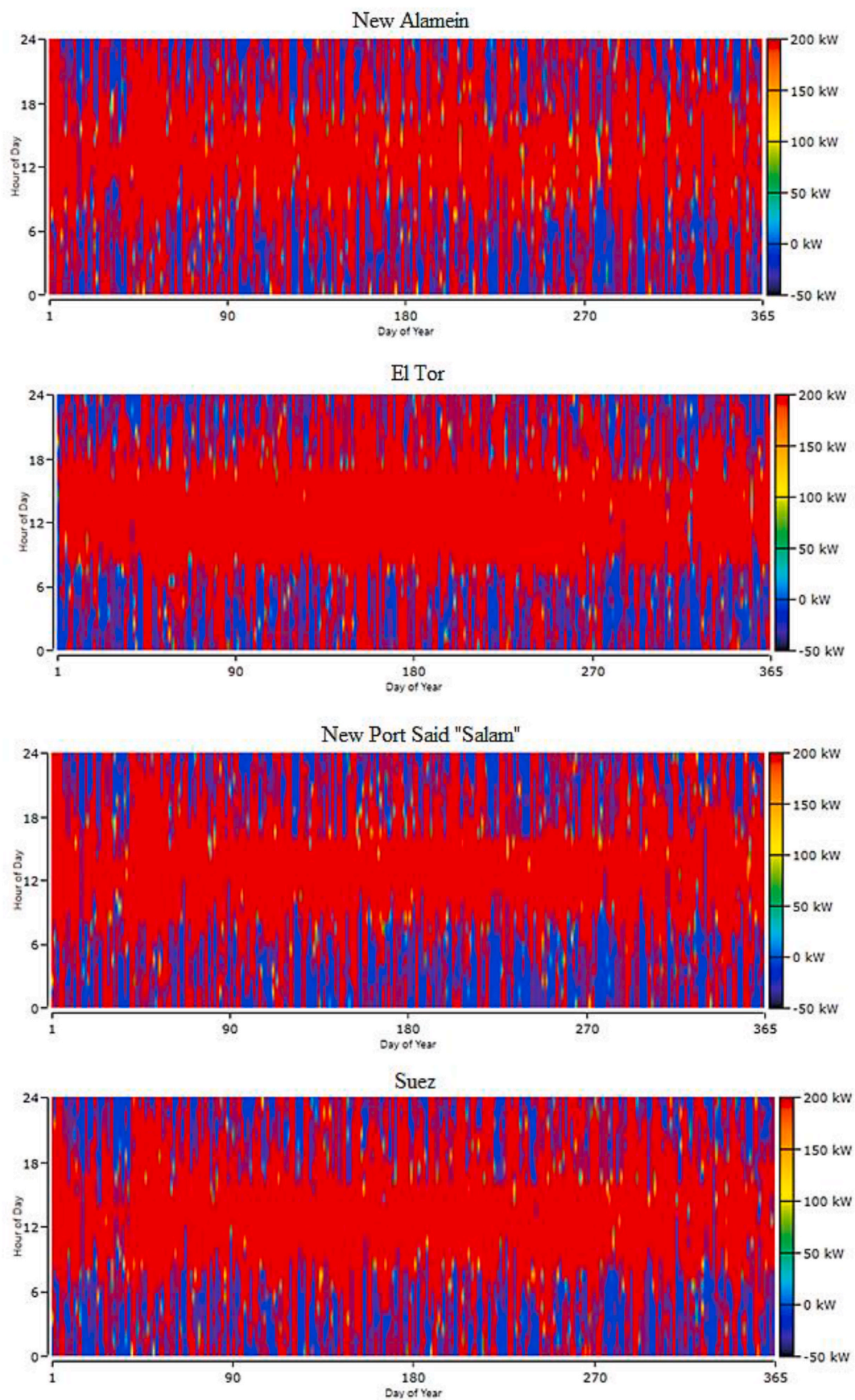


Fig. 14. Monthly energy sold to the grid for grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid system for the optimal case of all sites studied.

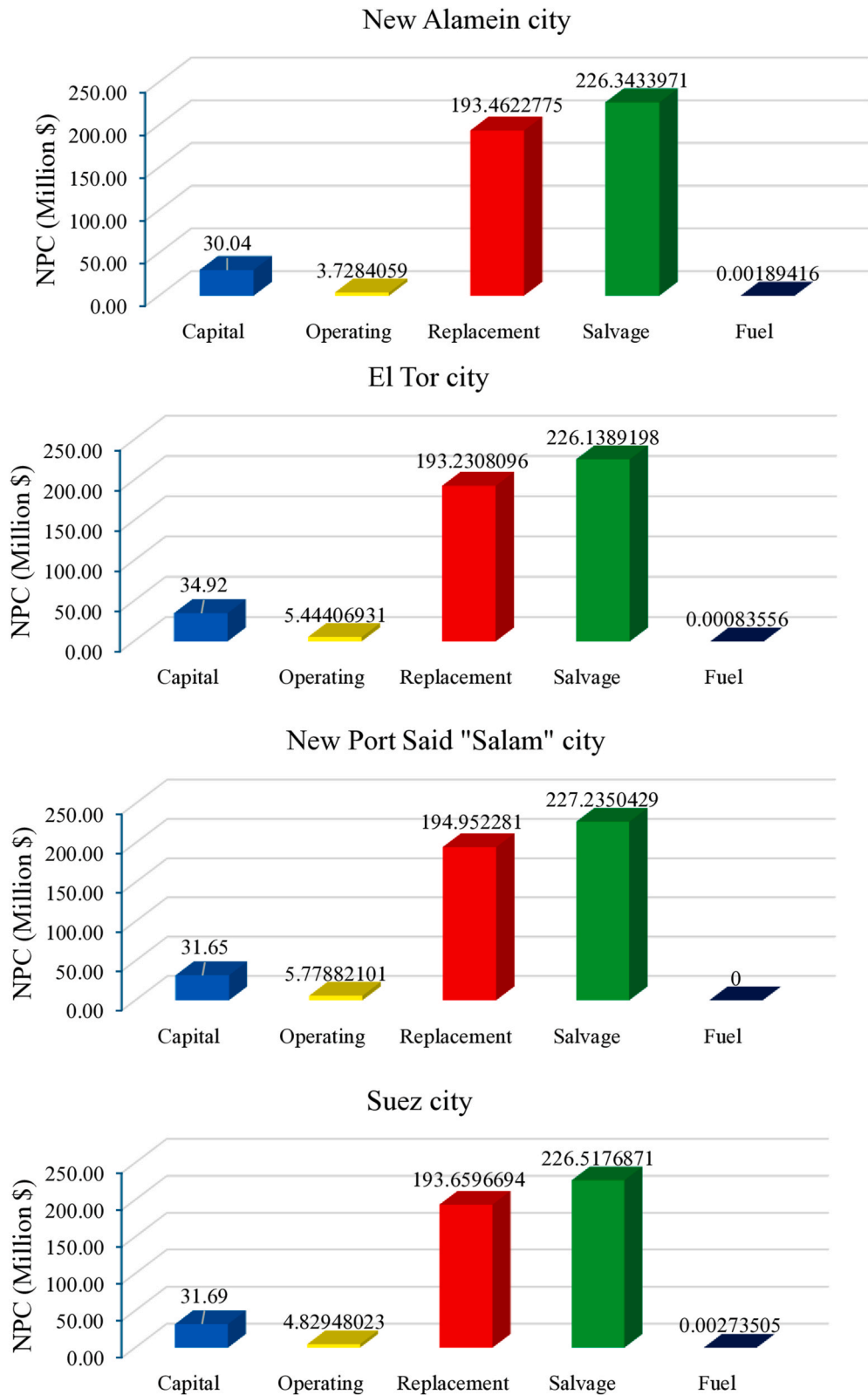


Fig. 15. Cost types comparison of grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid system for the optimal case of all sites studied.

essential to consider the operational role of the diesel generator within the proposed hybrid system. In the present study, the diesel generator contributes only a negligible fraction of the total energy production, ranging from 0.0000586 % to 0.000200 % as shown in Tables 7 and 13. This clearly demonstrates that the DG is not intended to serve as a

primary energy source, but rather as an emergency backup to guarantee supply continuity during rare instances of prolonged low renewable resource availability. Accordingly, the limited reliance on fossil fuels can be considered acceptable in the short to medium term, as it ensures operational reliability while Egypt continues to expand its renewable

Table 14

Sensitivity analysis of cost parameters for New Alamein City with 50 wind turbines under different discount and inflation rates.

	Real Discount rate %	Inflation rate %	Optimal Scenario	NPC \$M	LCOE \$/kWh	LCOH \$/kg
Case 1	7.5	2	3	24.14	0.7113	40
Case 2	10.5	8	4	25.77	0.5467	30.7
Case 3	12.5	14	4	27.96	0.3746	21
Case 4	15	20	4	29.05	0.2545	14.3
Case 5	18	26	4	27.2	0.1655	9.33
Case 6	21.25	33.7	4	0.89	0.0031	0.177

energy and storage infrastructure.

Furthermore, connecting these operational considerations with the overall system performance highlights the trade-off between cost and environmental impact. In parallel, the trade-off NPC and CO₂ emissions further underscores the practicality of the proposed design. As shown in Table 6, the overall system emissions remain significantly lower than those associated with conventional fossil-based generation. At the same time, the hybrid system achieves a substantial reduction in NPC, as highlighted in Tables 8–11 and Fig. 14, thereby demonstrating its economic robustness under Egypt's current conditions.

4.5.6. Sensitivity check for real discount and inflation rates

To evaluate the impact of economic assumptions on the techno-economic outcomes, a sensitivity analysis was performed by varying the real discount rate and inflation within representative ranges. The real discount rate was examined between 7.5 % and 21.25 %, while inflation was considered in the range of 2–33.7 %. The results indicate that both NPC and LCOH are highly sensitive to these parameters, as illustrated in Table 14. An increase in the real discount rate leads to a reduction in NPC, because higher discount rates lower the present value of future costs. This behavior is consistent with findings reported in previous studies and financial analyses [41]. Additionally, the ranking and relative feasibility of the proposed system configurations remain stable across the tested ranges, confirming the robustness of the techno-economic viability of the study's conclusions.

5. Conclusion and future outlook

Four scenarios for electricity and hydrogen production are studied in four Egyptian cities with different climates. Photovoltaics, WTs, batteries, PEME, H₂ tanks, and DGs are used as a standalone and grid-connected hybrid system. Three cases are studied for each scenario with 15, 20, and 50 WTs. This system was designed to produce about 121.45 kg/day of hydrogen with an average of 5.06 kg/h and 3443.7 kWh/day of electrical energy. This study is based on several factors such as energy analysis, hydrogen analysis, excess electricity, economic transactions, and environmental impact. The next step is to study the cost of transporting hydrogen within Egypt to remote areas via railways

or local natural gas network pipelines or outside Egypt for export overseas via ships. Homer Pro simulated more than 176,476 solutions. About 3268 of them are feasible and the rest are infeasible due to the capacity shortage constraint and maximum unmet hydrogen.

After the accurate simulation, the study can conclude that the grid-connected PV/WT/DG/Battery/PEME/H₂ tank hybrid system is the optimal configuration for hydrogen production in coastal cities in Egypt. The optimal system in New Alamein consists of 1107 kW solar panels, 2500 kW WTs, a 10 kW fixed capacity DG, 4 kWh Li-ion batteries, 1025 kW converters, and a 400 kg H₂ tank. The system NPC, LCOE, and LCOH are \$0.89 million, 0.00310 \$/kWh, and 0.177 \$/kg, respectively. The proposed system in El Tor consists of 3082 kW solar panels, 2500 kW WTs, a 10 kW fixed capacity DG, 4 kWh Li-ion batteries, 847 kW converters, and a 400 kg H₂ tank. The system NPC, LCOE, and LCOH are \$7.46 million, 0.0259 \$/kWh, and 1.48 \$/kg, respectively. The proposed system in New Port Siad "Salam" consists of 1686 kW solar panels, 2500 kW WTs, a 10 kW fixed capacity DG, 5 kWh Li-ion batteries, 1050 kW converters, and a 400 kg H₂ tank. The system NPC, LCOE, and LCOH are \$5.14 million, 0.0182 \$/kWh, and 1.02 \$/kg, respectively. The proposed system in Suez consists of 1748 kW solar panels, 2500 kW WTs, a 10 kW fixed capacity DG, 4 kWh Li-ion batteries, 1177 kW converters, and a 400 kg H₂ tank. The system NPC, LCOE, and LCOH are \$3.66 million, 0.0128 \$/kWh, and 0.728 \$/kg, respectively. Optimizing system configurations results in substantial cost reductions: in New Alamein, the net present cost and levelized cost of hydrogen decrease by up to 89.5 %, while the levelized cost of energy drops by 94.8 %. Comparable reductions are observed in El Tor (50 % and 75 %), New Port Said (58 % and 79 %), and Suez (67 % and 84 %), respectively. The proposed systems attain a 100 % renewable energy fraction, effectively eliminating greenhouse gas emissions. Surplus electricity is exported to the grid, contributing to a reduction in the system's net present cost. Notably, the systems operate independently of grid electricity purchases. Furthermore, the optimal configurations fully satisfy the entire hydrogen demand.

CRediT authorship contribution statement

Mohamed Osman Atallah: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Abdallah M. Elsayed:** Writing – original draft, Validation, Investigation, Data curation, Conceptualization. **Mohammed H. Alqahtani:** Writing – review & editing, Visualization, Investigation, Formal analysis. **Abdullah M. Shaheen:** Writing – review & editing, Supervision, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Load description

Table A1 displays the electrical loads for the different apartment areas utilized in this study, Also, Table A2 and A3 illustrate the estimated daily load and energy consumption, respectively, for each apartment model during the different seasons. Additionally, Table A4 tabulates the overall load characteristics while Table A5 shows the total estimated daily hydrogen and electric loads for the proposed site during the different seasons.

Table A.1

Electrical loads for the different apartment areas [23].

Load	110 m ² apartment				130 m ² apartment				150 m ² apartment			
	No.	Power (W)	Usage h/day	Energy Wh/day	No.	Power (W)	Usage h/day	Energy Wh/day	No.	Power (W)	Usage h/day	Energy Wh/day
Basic light	14	18	2	504	16	18	2	576	18	18	2	648
Decoration light	6	9	1	54	8	9	1	72	10	9	1	90
Small appliances												
Wall fan	2	65	8	1040	3	65	4	780	3	65	4	780
Stand fan	1	60	8	480	2	60	4	480	2	60	4	480
Kitchen fan	1	40	3	120	1	40	3	120	1	40	3	120
Bathroom fan	1	36	1	36	2	36	1	72	3	36	1	108
Electric hair clipper	–	–	–	–	1	10	0.1	1	1	10	0.1	1
Essential appliances												
Refrigerator	1	400	7.2	2880	1	500	7.2	3600	1	700	7.2	5040
Freezer	1	90	7.2	648	1	150	7.2	1080	1	310	7.2	2232
Power appliances												
Water heater	1	800	0.5	400	2	900	0.5	900	3	1100	0.5	1650
Air conditioning	1	900	3	2700	1	900	4	3600	2	1200	5	1200
Electric oven	–	–	–	–	–	–	–	–	1	3000	1	3000
Microwave oven	–	–	–	–	1	700	0.2	140	1	800	0.2	160
Fan heater	1	1500	3	4500	1	1500	3	4500	1	1600	2	3200
Oil heater	–	–	–	–	–	–	–	–	1	1700	3	5100
Laundry	1	500	0.25	125	1	700	0.25	175	2	700	0.15	210
Kettle	–	–	–	–	1	1100	0.3	330	1	1200	0.2	240
Coffee maker	–	–	–	–	–	–	–	–	1	700	0.1	70
Food blender	1	400	0.25	100	1	500	0.25	125	1	600	0.25	150
Iron	1	800	0.1	80	1	1100	0.1	110	1	1400	0.1	140
Vacuum machine	1	800	0.25	200	1	1500	0.25	375	1	2000	0.15	300
Hair dryer	–	–	–	–	–	–	–	–	1	1000	0.1	100
Media appliances												
Radio	1	5	2	10	–	–	–	–	–	–	–	–
Television	1	57	3	171	2	65	2	260	3	72	1	216
Personal computer	1	135	3	405	2	150	2	600	4	150	1	600

Table A.2

The estimated daily load for each apartment model during the different seasons

Time	Spring/Summer			Fall/Winter		
	110 m ²	130 m ²	150 m ²	110 m ²	130 m ²	150 m ²
12 : 05 a.m.	545.167	654	1469.5	193.5	249	364.5
05 : 06 a.m.	745.167	1104	2294.5	393.5	699	1189.5
06 : 07 a.m.	197	492.5	2113	197	492.5	2113
07 : 04 p.m.	147	195	303	147	195	303
04 : 06 p.m.	347	645	1128	347	645	1128
06 : 07 p.m.	647.167	1015.83	3355.83	295.5	610.83	2250.83
07 : 08 p.m.	1002.167	1378.33	2295.83	650.5	973.33	1190.83
08 : 09 p.m.	792.497	1004.997	1817.83	1940.83	2099.997	3479.497
09 : 11 p.m.	740.497	740.667	1741.5	1888.83	2035.667	3403.167
11 : 12 a.m.	545.167	654	1469.5	193.5	249	364.5

Table A.3

The daily energy consumption for each building during the different seasons

Area/Season	Spring/Summer	Fall/Winter
110 m ²	99530	102330
130 m ²	133960	130360
150 m ²	113340	93500
Total demand (Wh/day)	346830	326190

Table A.4
Overall load characteristics

Annual electric usage	1.257	GWh/yr
Hourly peak load	926.56	kW
Hourly average load	143.49	kW
Load factor	0.15	–

Table A.5
The total estimated daily hydrogen and electric loads for the proposed site during the different seasons

Time	Spring/Summer		Fall/Winter	
	Hydrogen (kg/hr)	Electric (kW/hr)	Hydrogen (kg/hr)	Electric (kW/hr)
12 : 05 a.m.	6.301	178.677	2.075	58.830
05 : 06 a.m.	9.758	276.697	5.531	156.830
06 : 07 a.m.	5.412	153.470	5.412	153.470
07 : 04 p.m.	1.634	46.320	1.634	46.320
04 : 06 p.m.	5.09	144.320	5.09	144.320
06 : 07 p.m.	10.599	300.533	6.371	180.666
07 : 08 p.m.	11.634	329.883	7.406	210.016
08 : 09 p.m.	8.903	252.462	19.159	543.262
09 : 11 p.m.	7.680	217.776	18.641	528.576
11 : 12 a.m.	6.301	178.677	2.075	58.830

Appendix B. Wind Speed Height Adjustment

To estimate wind speeds at the hub heights of the selected wind turbines, HOMER Pro applies the logarithmic wind profile law [29]:

$$V(z) = V(Z_{ref}) \times \frac{\ln \frac{z}{z_0}}{\ln \frac{Z_{ref}}{z_0}}$$

where:

- $V(z)$ is the wind speed at the desired hub height z (m),
- $V(Z_{ref})$ is the reference wind speed at height Z_{ref} (50 m in NASA POWER data),
- z_0 is the surface roughness length, which depends on terrain characteristics.

For coastal regions such as the study sites, HOMER assumes a typical surface roughness length of 0.03 m, corresponding to open flat terrain with low vegetation. This adjustment provides a reliable estimation of wind speeds at turbine hub heights (80, 100) m, ensuring consistency in the performance evaluation of different wind turbine scenarios. The resulting wind speed adjustment curves for each hub height are shown in Fig. B1.

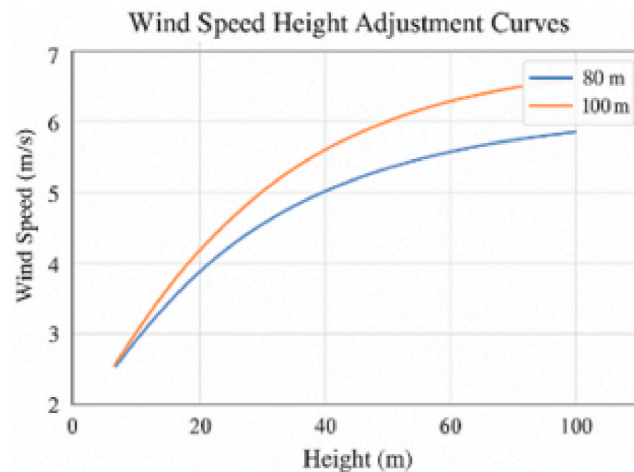


Fig. B1. Height adjustment curves for wind speed based on NASA POWER reference data at 50 m, extrapolated to hub heights of 80 m and 100 m using the logarithmic profile law with a surface roughness length of 0.03 m (coastal terrain).

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